North American Bat Monitoring Program in SE Alaska

Data Summary and Activity Trend Analyses: 2016 - 2025

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within S.E Alaska, with key support from state and provincial representatives. In 2025, 11 grid cells in S.E. Alaska were monitored. Results indicate that bat populations across S.E. Alaska remain stable, with some key species showing small positive trends, including Little Brown Myotis and Eastern Red Bat. Given growing threats to bat populations in BC, continued monitoring is essential to detect any significant changes.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in S.E Alaska, Alberta, and British Columbia.

NNW Bat Hub was established in 2024 with collaboration from Government of Alaska, Government of Alberta, and Government of British Columbia. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska Hub. The implementation of the program within S.E Alaska has key support from Alaska Department of Fish and Game.

2.2 Threats to bat populations in Alaska

Bat populations in Alaska face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

2.2.1 White-Nose Syndrome

White-nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in 42 U.S. states, nine Canadian provinces, and the Northwest Territories (U.S. Fish and Widlife Service (USFWS), n.d.-a; Segers, n.d.) . *Pd* , but not WNS, has been detected in 5 additional states and within British Columbia(Figure 1: Segers (n.d.) ; U.S. Fish and Widlife Service (USFWS) (n.d.-a)). Continued monitoring of populations in Alaska is critical as *Pd* and WNS continue to spread throughout the region.

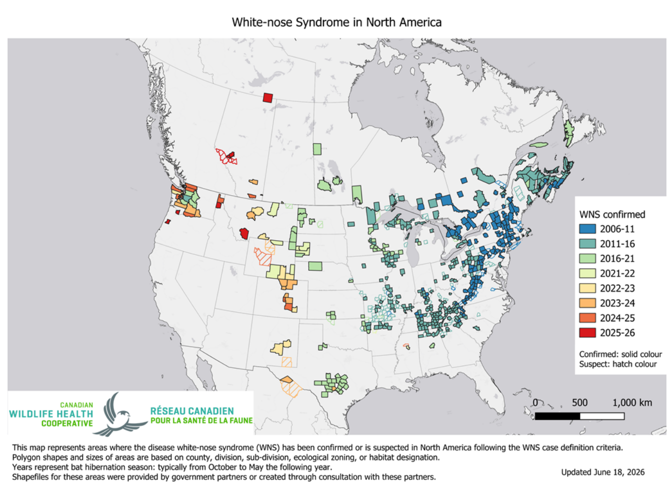


Figure 1: White-nose syndrome (WNS) and Pseudogymnoascus destructans (Pd) occurrence in The US and Canada. Source: CWHC-RCSF (2026)

2.2.2 Wind Energy Development

Wind energy development is a substantial source of bat mortality in North America; one 2023 estimate placed U.S. turbine-related deaths at 926,856 (*NASBR Statement on Wind Energy Impacts on Bat Populations* 2024). Migratory tree-roosting species bear the brunt, accounting for roughly 73% of reported carcasses nationally — chiefly the hoary bat (Lasiurus cinereus), eastern red bat (Lasiurus borealis), and silver-haired bat (Lasionycteris noctivagans) (Zimmerling and Francis 2016). Two of these, the silver-haired and hoary bats, occur in Southeast Alaska. Modeling suggests turbine collisions alone could reduce the North American hoary bat population by up to 90% over 50 years (Frick et al. 2017). Wind generation in Alaska is concentrated along the Railbelt, on Kodiak Island, and in rural western and northern communities, with little development in the Southeast to date (Hoen et al., n.d.). As renewable energy expands, however, future projects would pose a collision risk to the region's migratory bats, whose distribution and seasonal movements there remain poorly understood.

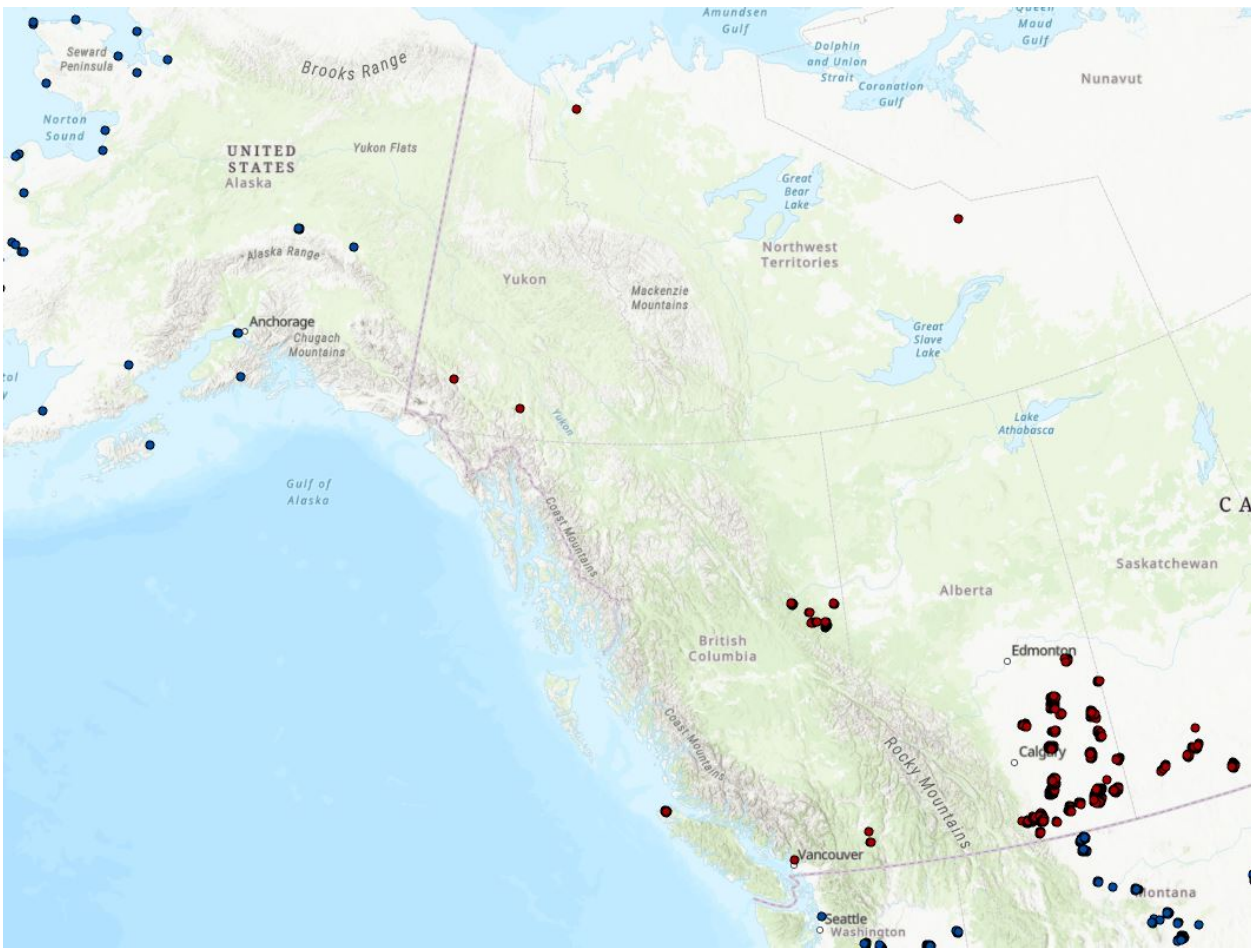


Figure 2: Current wind turbines across the region. Data from Canadian Wind Turbine database and U.S. Wind Turbine database (Hoen et al., n.d.; Natural Resources Canada, n.d.).

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Unlike birds, bats rely on echolocation rather than song, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local assemblage.

2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.

3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.

4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

2.4 Update in Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

3 Methods

3.1 Data Collection

3.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites are selected to maximize species detection and recording quality across different habitat types (see (Rae and Lausen 2024) for full site selection criteria). Deployment occurs between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing are kept consistent between years to minimize variability in conditions affecting bat activity.

3.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Two replicates of a 30–35 km route are driven at 30–35 km/h with a rooftop detector, using straight roads that minimize switchbacks. Transects are conducted concurrently with stationary deployments.

3.1.3 Model Covariates

Dynamic occupancy model covariates were based on those established by Rae and Lausen (2024) , with additional exploration of factors influencing colonization and extinction. Specifically we included distance to roads derived from the unpublished Human Footprint Index (HFI) layer currently being developed by the Geospatial Centre within Biodiversity Pathways.

Where possible, temperature and relative humidity are recorded in each grid cell using onsite logging devices; otherwise, data are sourced from the nearest climate station. Additional covariates include nightly mean wind speed (km/h), cumulative precipitation since January 1 (mm), degree days above 10°C since January 1, and moon illumination (proportion illuminated × proportion of night above horizon). Site-level covariates — including distance to clutter, percent clutter, distance to water, and water feature type — are recorded by grid leaders during deployment.

3.1.4 Manual Verification

In 2025 acoustic analysis mostly followed the standards set by WCSC in previous years (Rae and Lausen 2024). This included running all files through two automatic classifiers, Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Northwestern British Columbia classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018) and previous analysis standards developed by WCSC. The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually verified. Instead, based on findings from Stratton et al. (2022) , at least 50% of recordings were selected for manual verification.

Subsetting of for manual verification

Starting in 2024, the NNW Bat Hub implemented manual vetting of only a subset of all the bioacoustic data collected under the NABat program. Given the project's large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and Brian Paterson, developed procedures to ensure that selecting a subset of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of tagged recordings at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years' identifications) and commonly misidentified species were fully vetted. Specifically, all Eastern Red Bat (*Lasiurus borealis*) tags were manually reviewed due to frequent misclassification as Little Brown Bats (*Myotis lucifugus*). Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and "NoID" recordings were randomly selected until 50% of all recordings were reviewed.

3.2 Statistical Analyses

3.2.1 Activity Trends

Statistical analyses followed closely what was established by WCSC and Dr. Carl Schwarz in 2021 (Rae and Lausen 2024). This includes conducting species-specific trend analyses examining activity patterns in the data collected to date using the (negative binomial GLMM) models and the covariates identified in **?@tbl-cov**.

Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years. We conducted our statistical analyses on provincial and regional scales. For the regional analyses, we assigned all sampled grid cells into five geographic regions (Alaska, Southwestern, Southcentral, Southeastern, and Northern), each roughly corresponding to different species sets. Since transects are only conducted in roughly 50% of our grid cells on a typical year and each transect records fewer bats per night than stationary monitoring, we have elected not to conduct regional-scale transect trend analyses due to low sample sizes.

3.2.1.1 Use of Automated Identification Results in Stationary Trend Analyses

Stationary trend analyses used AutoID classifications from Kaleidoscope Pro and Sonobat 3.0. Manual verification was applied only to flag non-bat noise; where manual review suggested a different species than the AutoID label, the original AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2024), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

3.2.2 Dynamic Occupancy

This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at GRTS-selected sites in British Columbia (BC) as part of the NABat long-term monitoring program. Data were filtered to the 2017–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using max 4 visits per site-year. Sites had to have a minimum of 2 visits within a given year and data from at least 2 years total. Models were run separately for all 17 bat species.

Fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. Initial occupancy (ψ) was modeled as a function of elevation and water nearby, with a random effect for region. Colonization (γ) and persistence (φ) were modeled with time-varying intercepts and distance to road as a covariate. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Juli

Elevation for each site was extracted from AWS terrain tiles (~30 m resolution, SRTM-based) using the elevatr package in R, distance to road and distance to harvest was extracted using the Biodiversity Pathways Human Footprint Index (not published yet). All continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

Models were fit in a Bayesian framework using JAGS via the jagsUI package in R, with 30,000 iterations, 15,000 burn-in, thinning by 5, and 3 chains run in parallel, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normally distributed (mean = 0, precision = 0.368, equivalent to SD   1.65), and the region-level random effect standard deviation received a uniform(0, 5) prior. Model convergence was assessed using Rhat diagnostics. Results are interpreted using posterior means and 95% Bayesian credible intervals. An overall occupancy trend metric ( ) was calculated as the ratio of estimated occupancy in the final year to occupancy in the initial year; values of > 1 with 95 CIs entirely above 1 indicate an increase in occupancy, values < 1 with 95 CIs entirely below 1 indicate a decline, and values with CIs overlapping 1 indicate uncertain or stable trends.

4 Results

In 2025 a total of 6 grid sites were sampled in SE Alaska and 3 newly established sites were sampled around the Anchorage area (Figure 3).

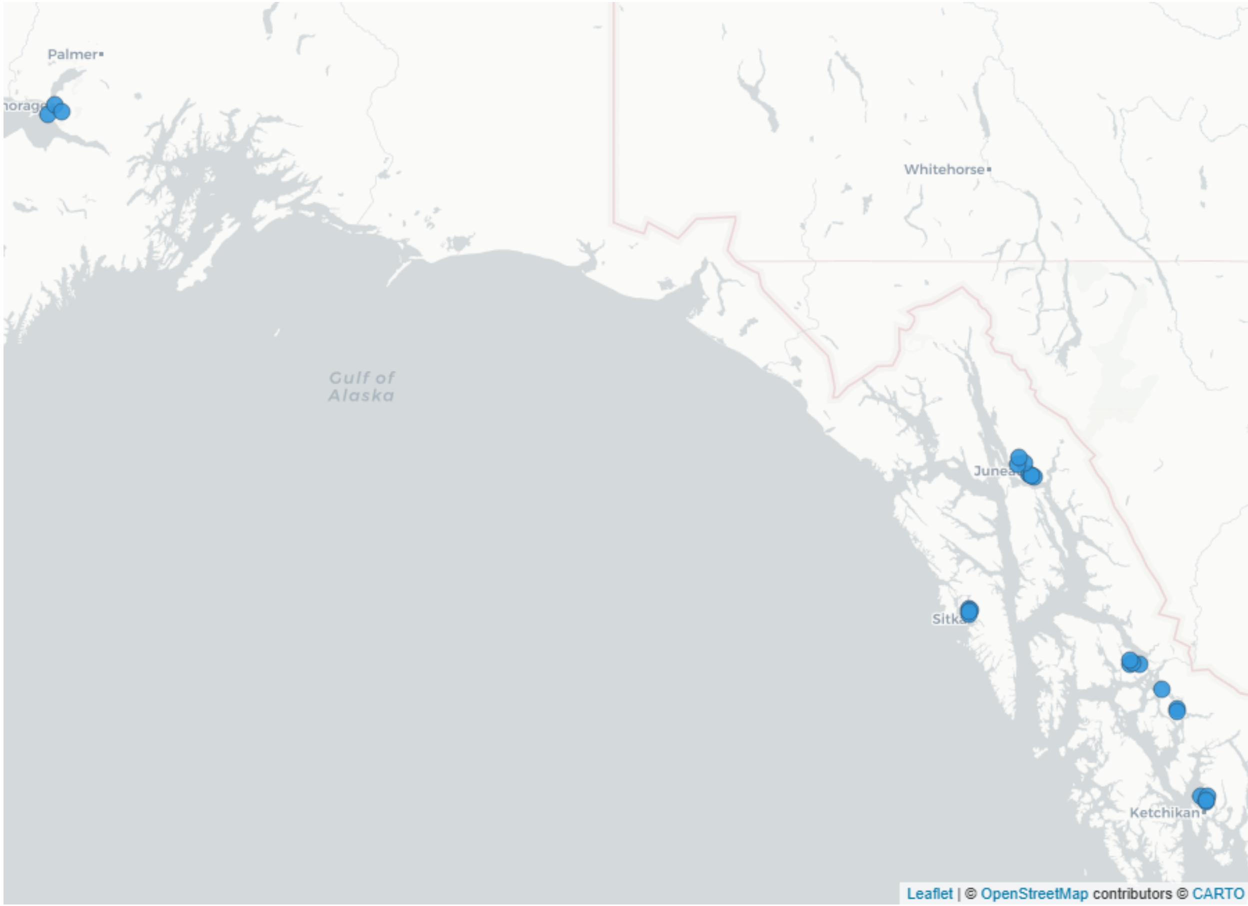


Figure 3: Acoustic bat monitoring sites in Alaska in 2025. Stationary recording stations are shown as blue circular markers

4.1 Activity Trends

Coming soon...

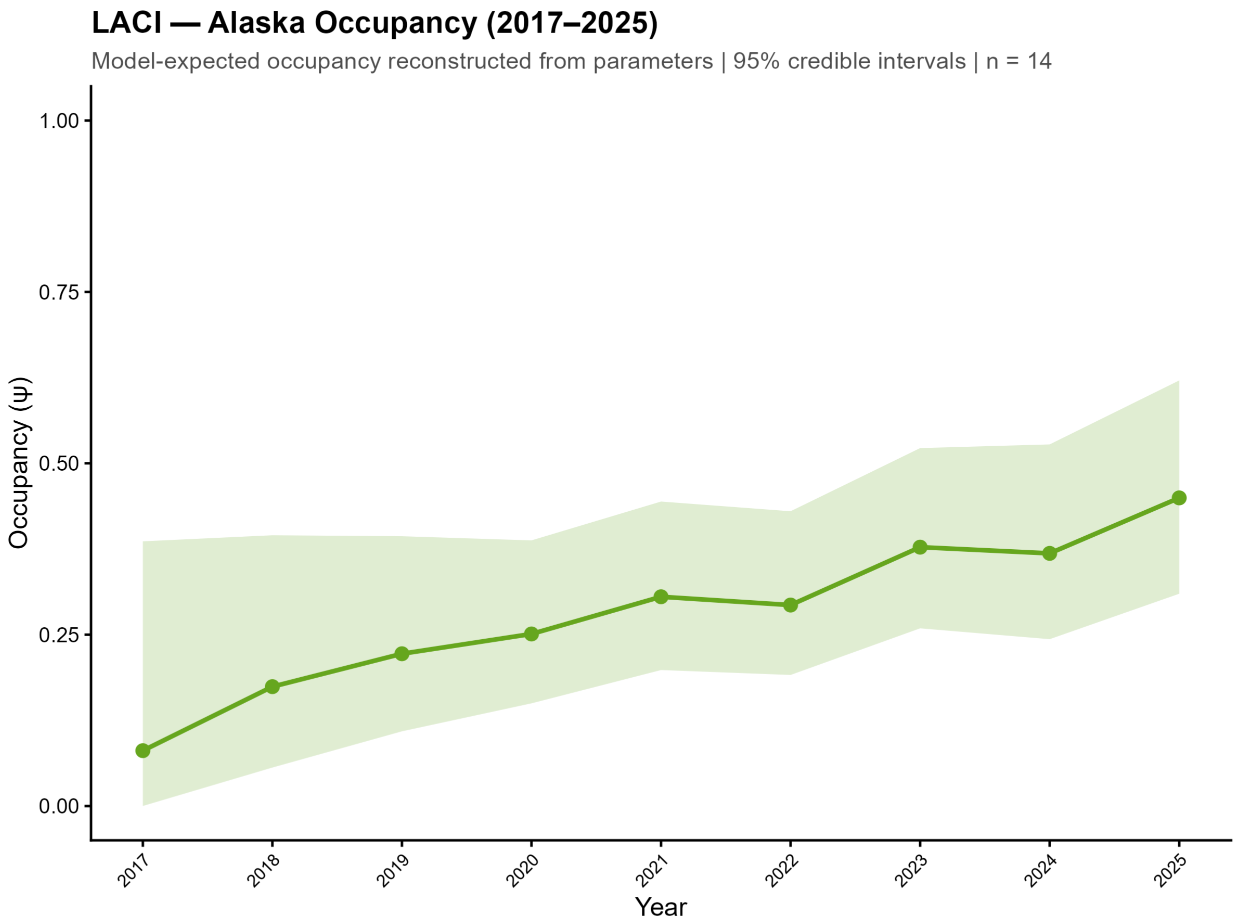
4.2 Dynamic Occupancy

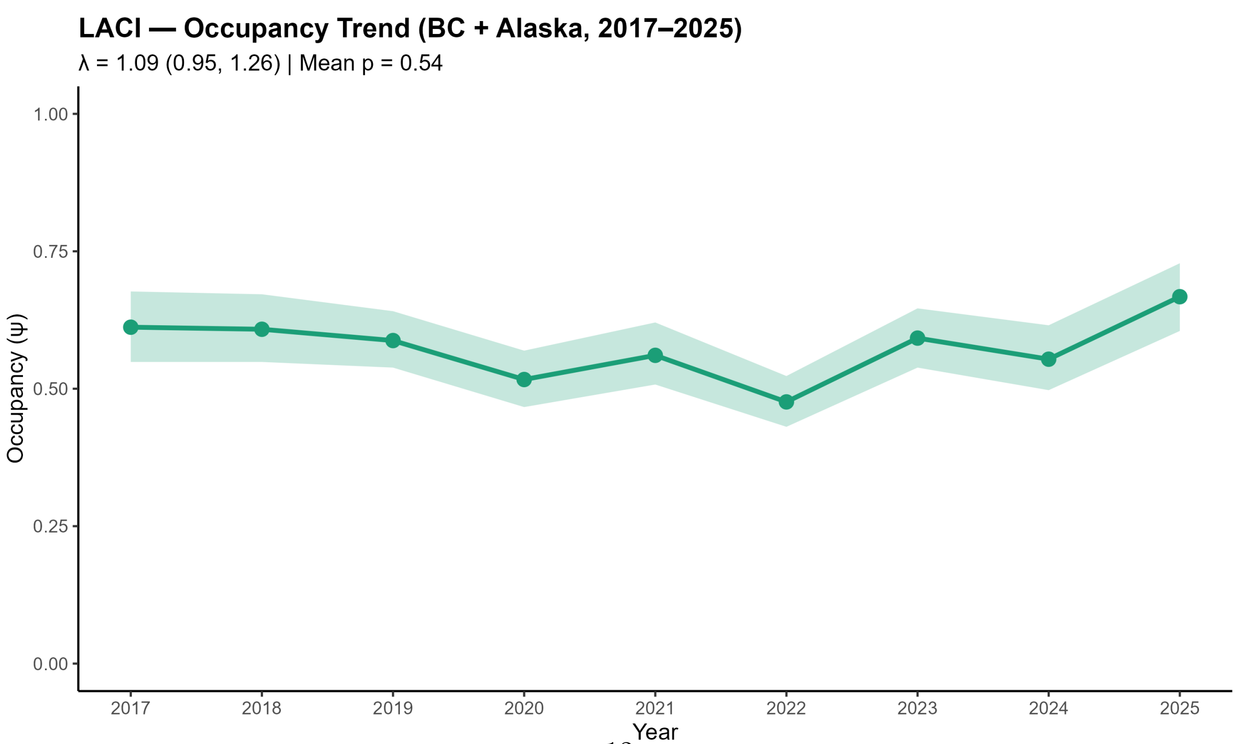
Temperature was the most consistent detection covariate–warmer nights equate to higher detection for most species. There were some clutter effects on detection probability that were species-specific. Julian date effects were mixed across species.

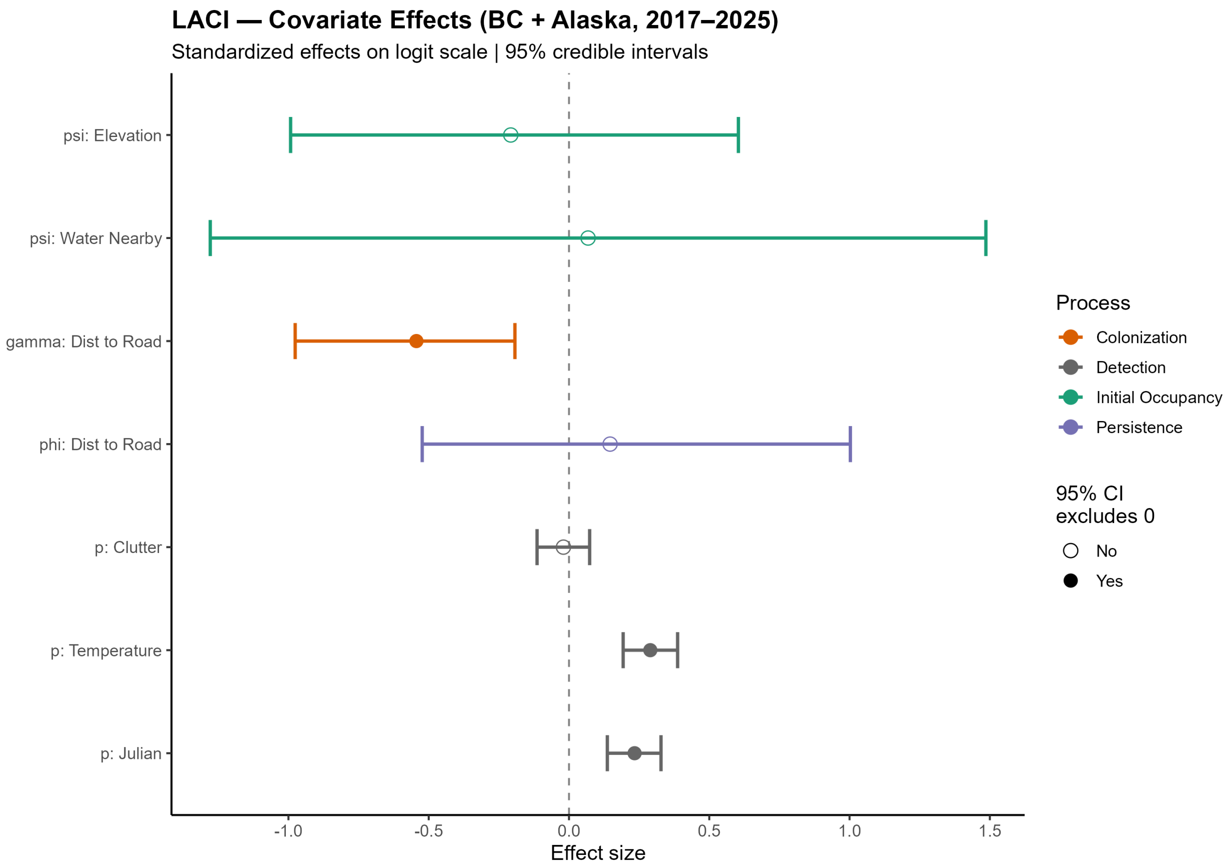
Species specific results can be found on Appendix A

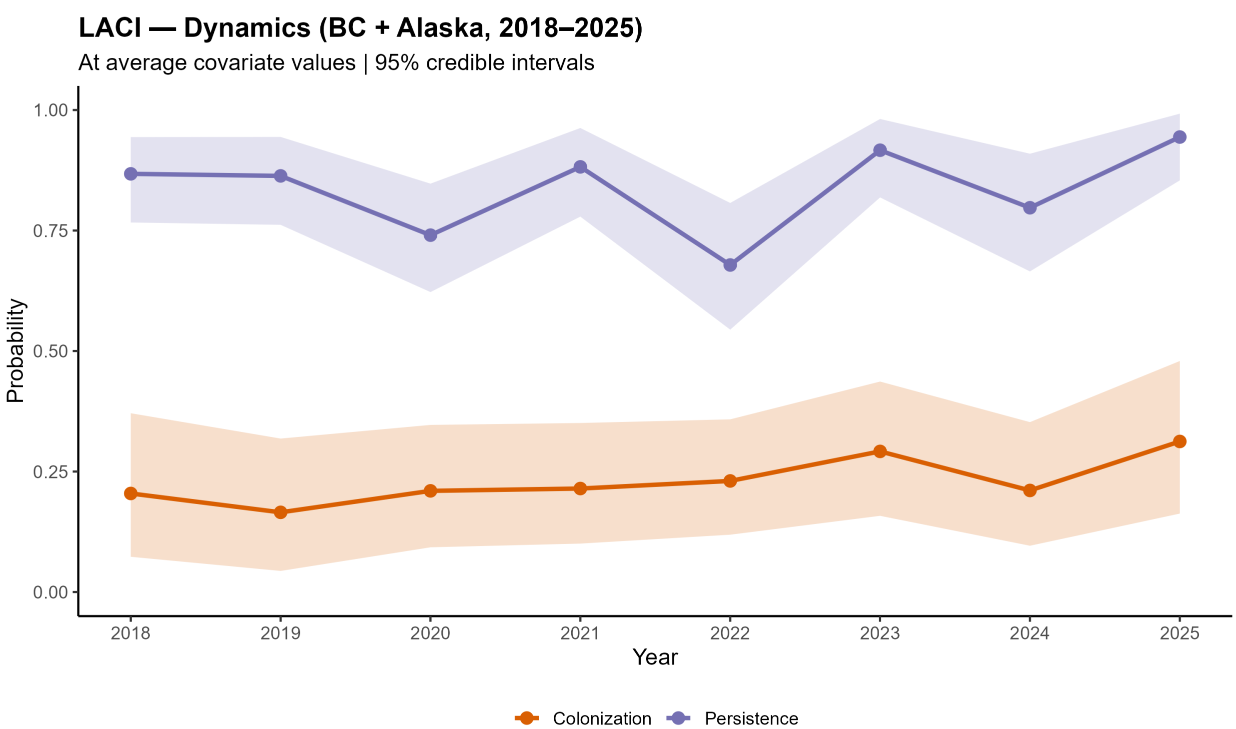
5 Appendix A

5.1 Hoary Bat (*Lasiurus cinereus*)

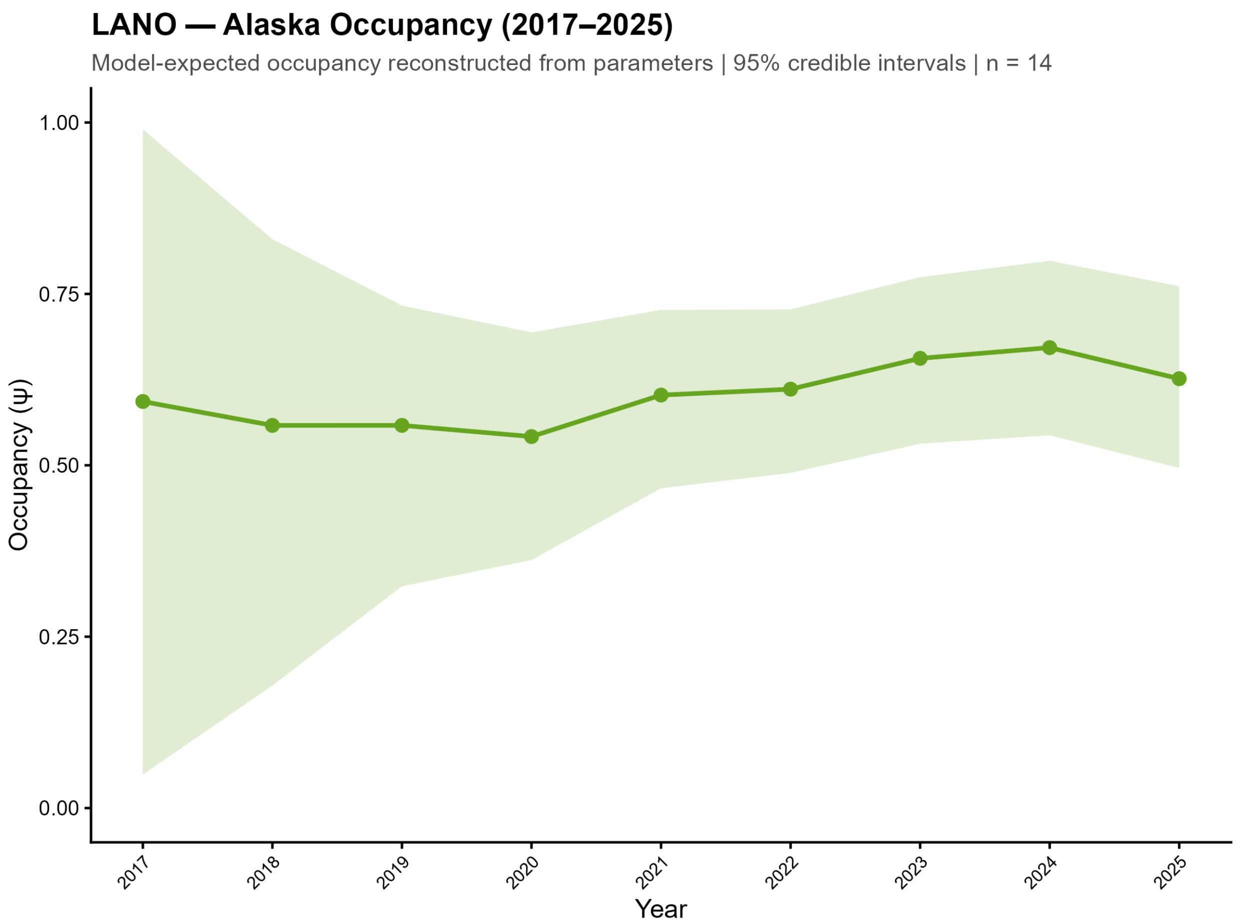


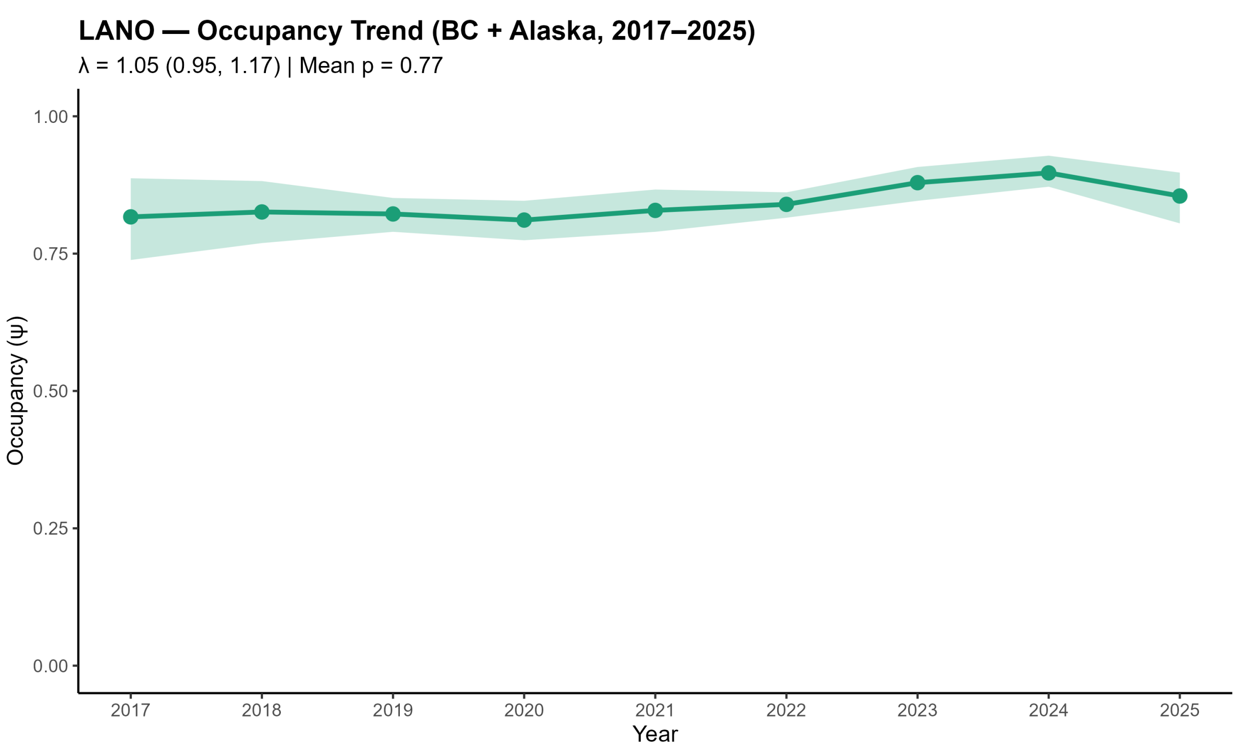


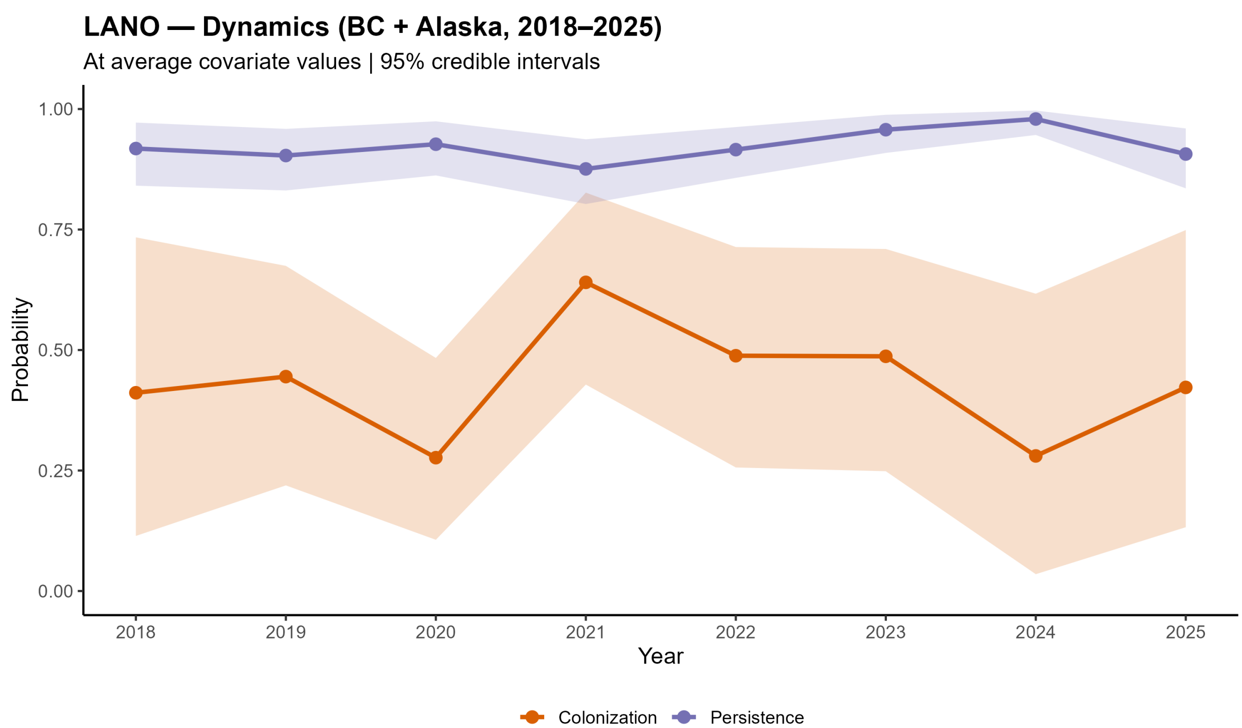




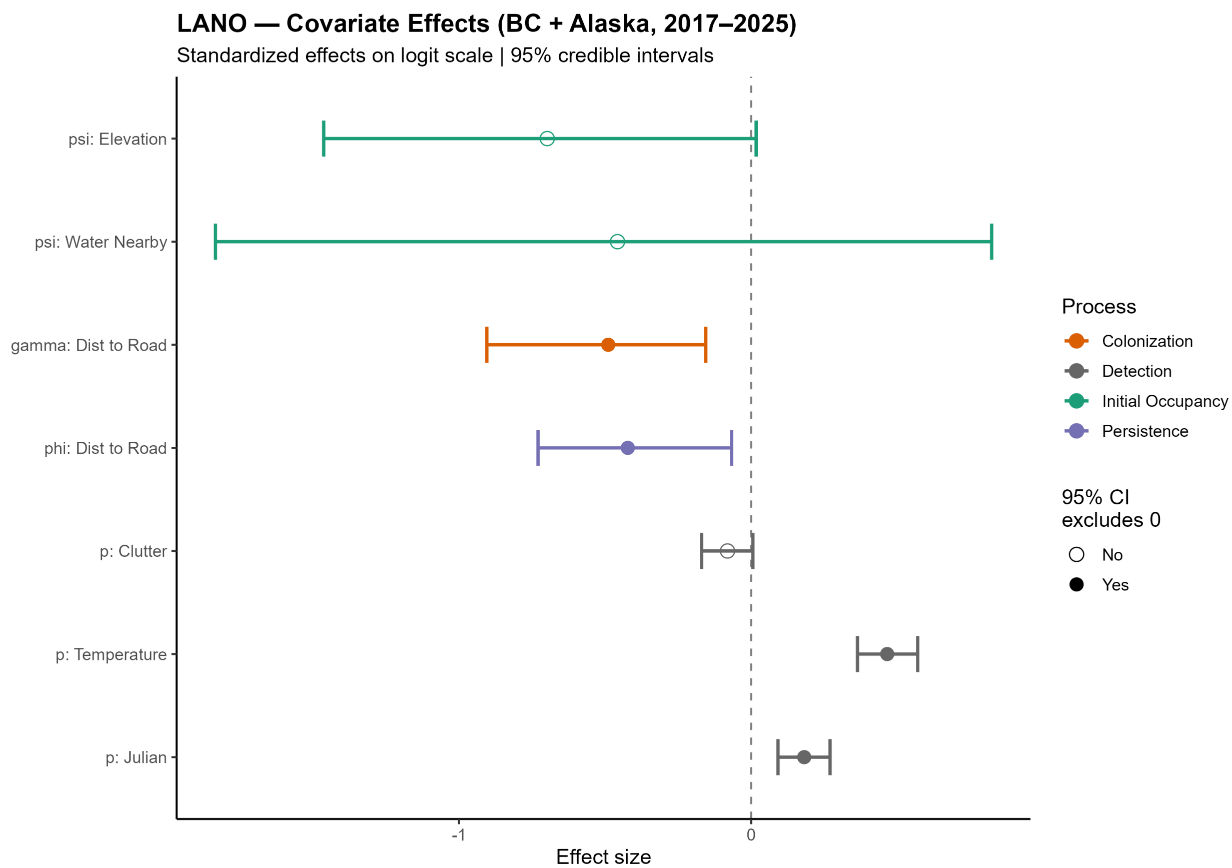
##Silver-haired Bat (*Lasyonicteris noctivagans*)


LANO — Alaska Occupancy (2017–2025)
Model-expected occupancy reconstructed from parameters | 95% credible intervals | n = 14
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


LANO — Occupancy Trend (BC + Alaska, 2017–2025)
λ = 1.05 (0.95, 1.17) | Mean p = 0.77
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

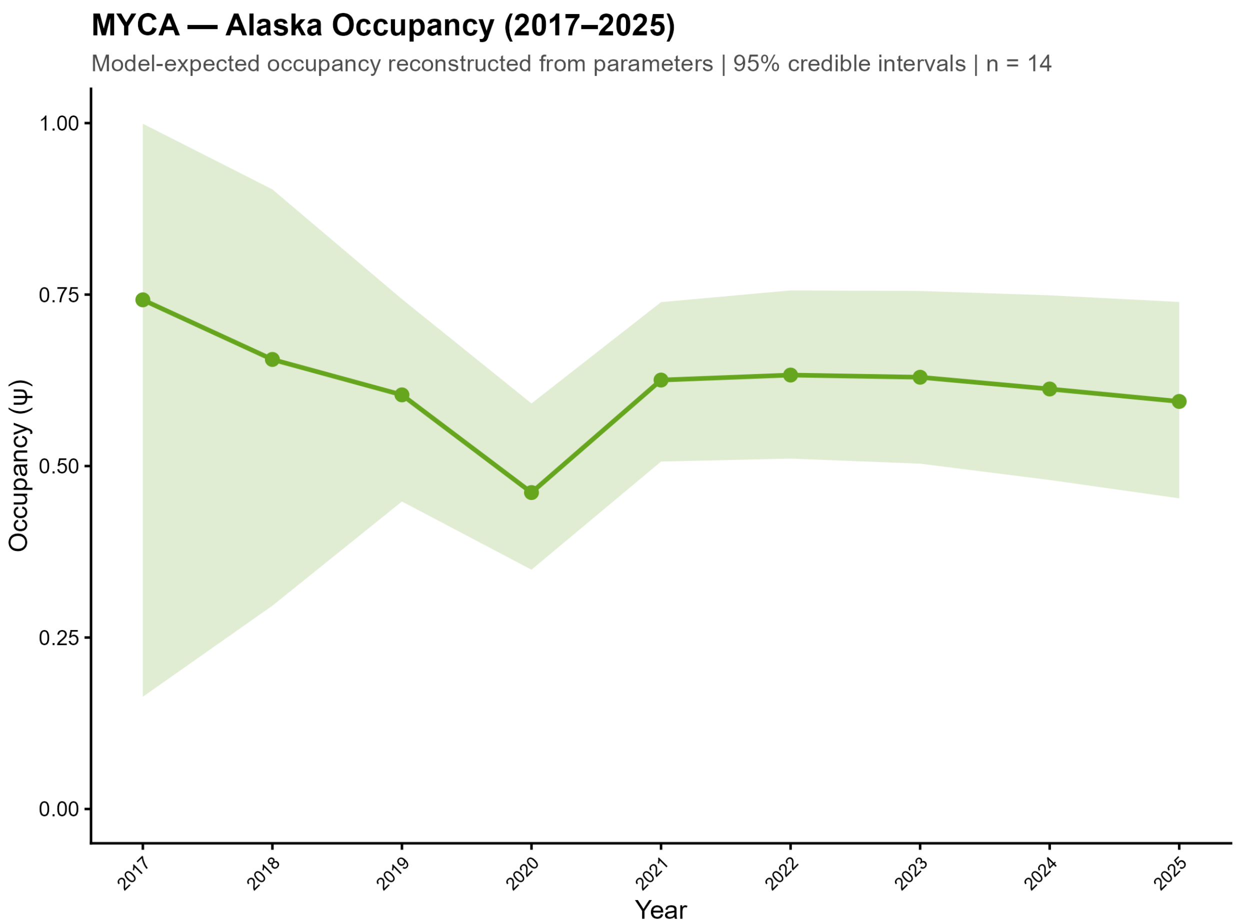


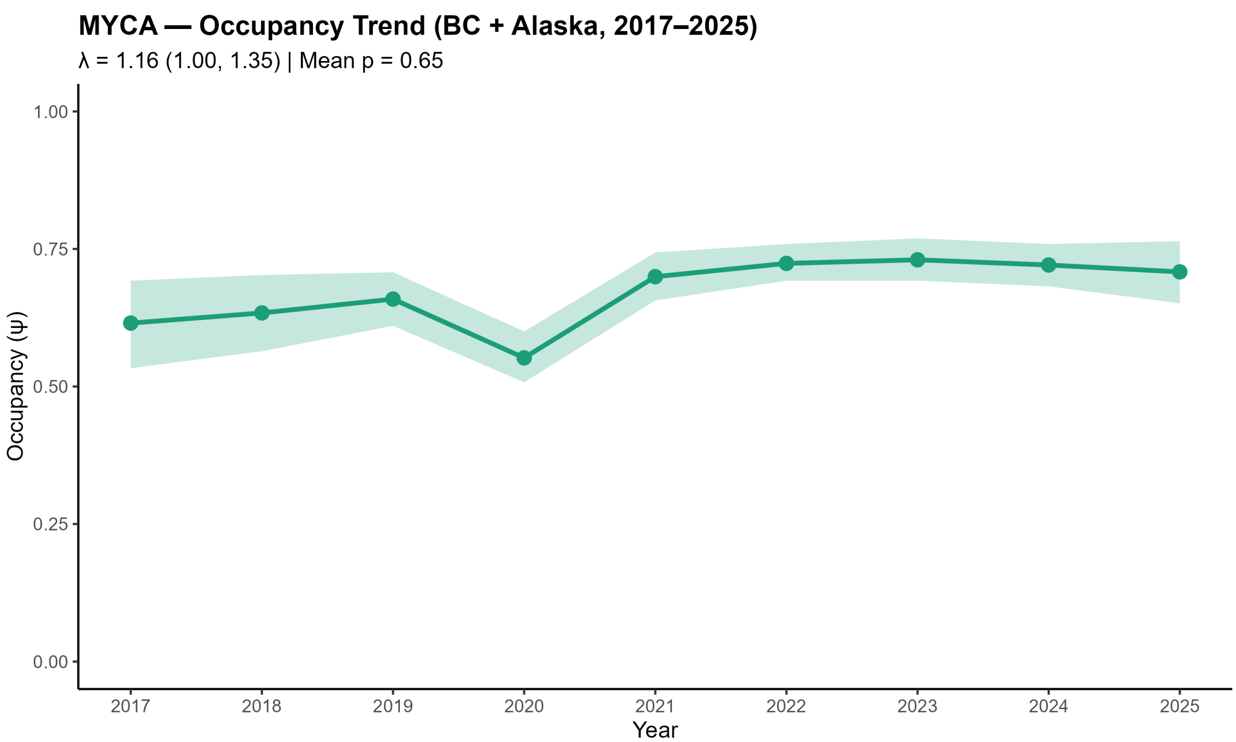
LANO — Dynamics (BC + Alaska, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

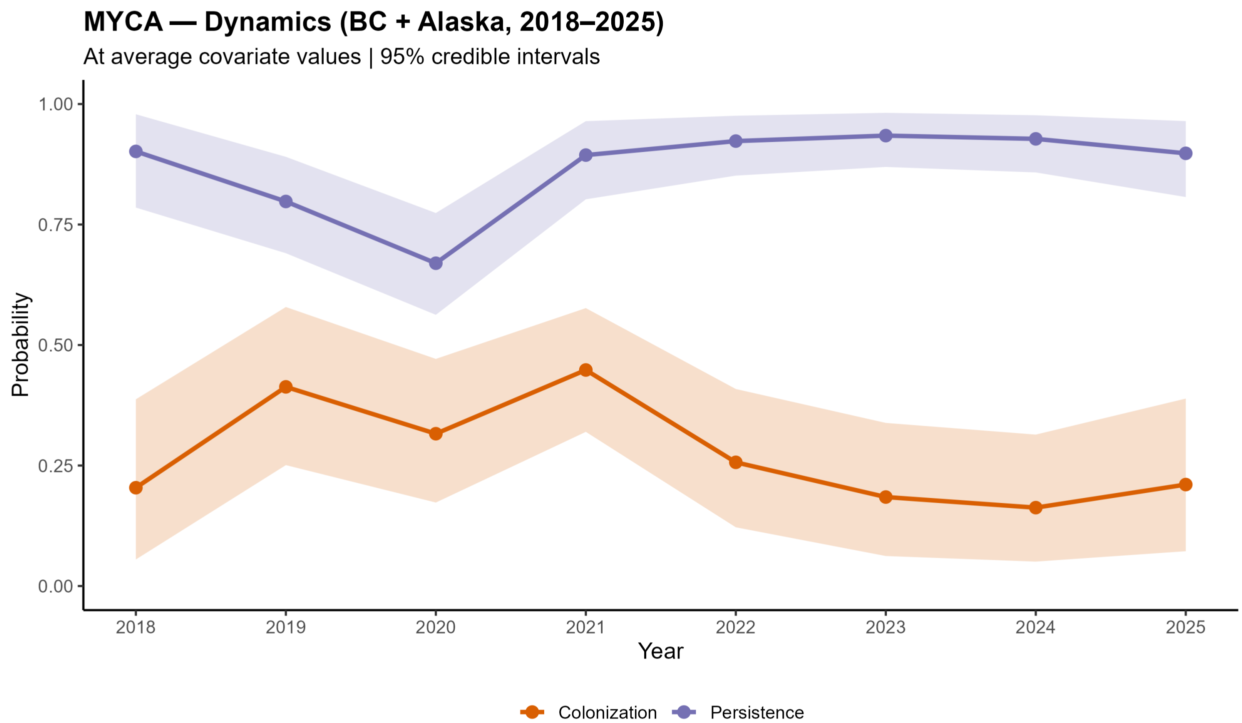


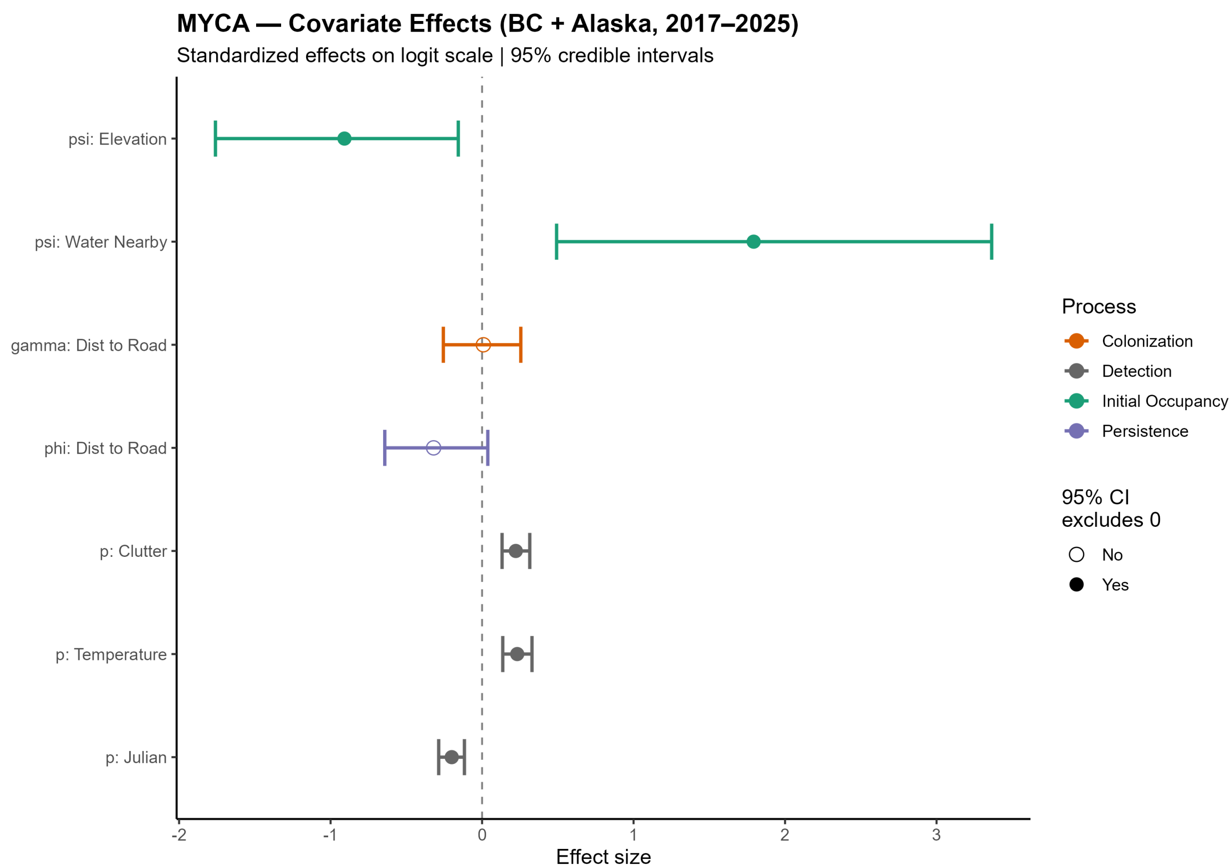
LANO — Covariate Effects (BC + Alaska, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
phi: Dist to Road
p: Clutter
p: Temperature
p: Julian
-1
0
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

5.2 California Myotis (*Myotis californicus*)

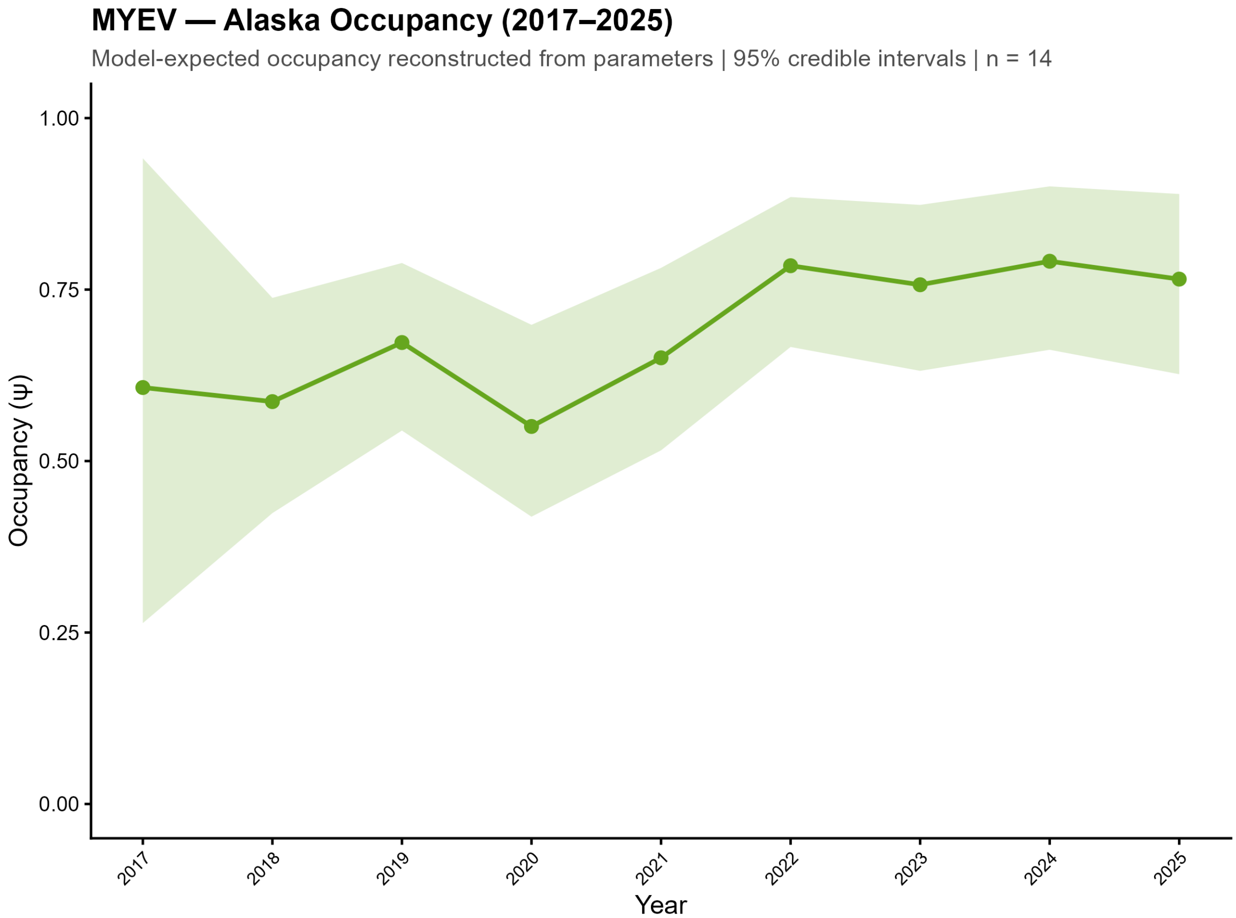


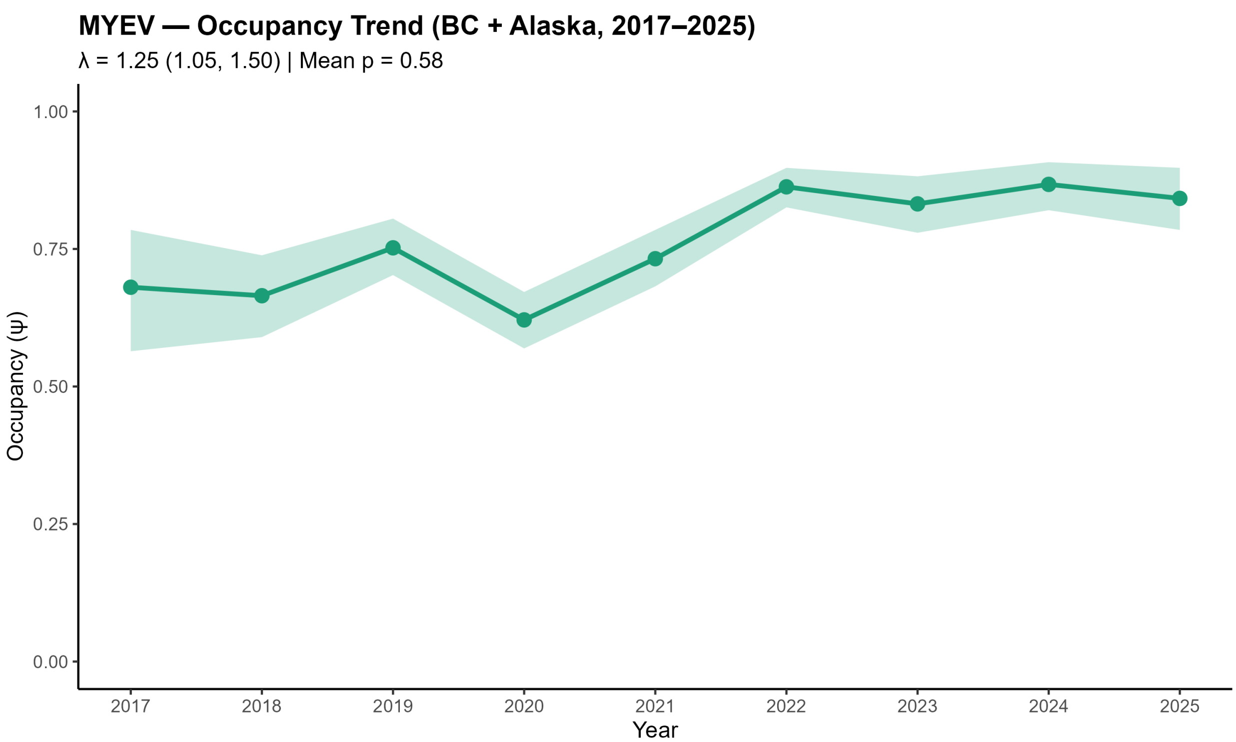


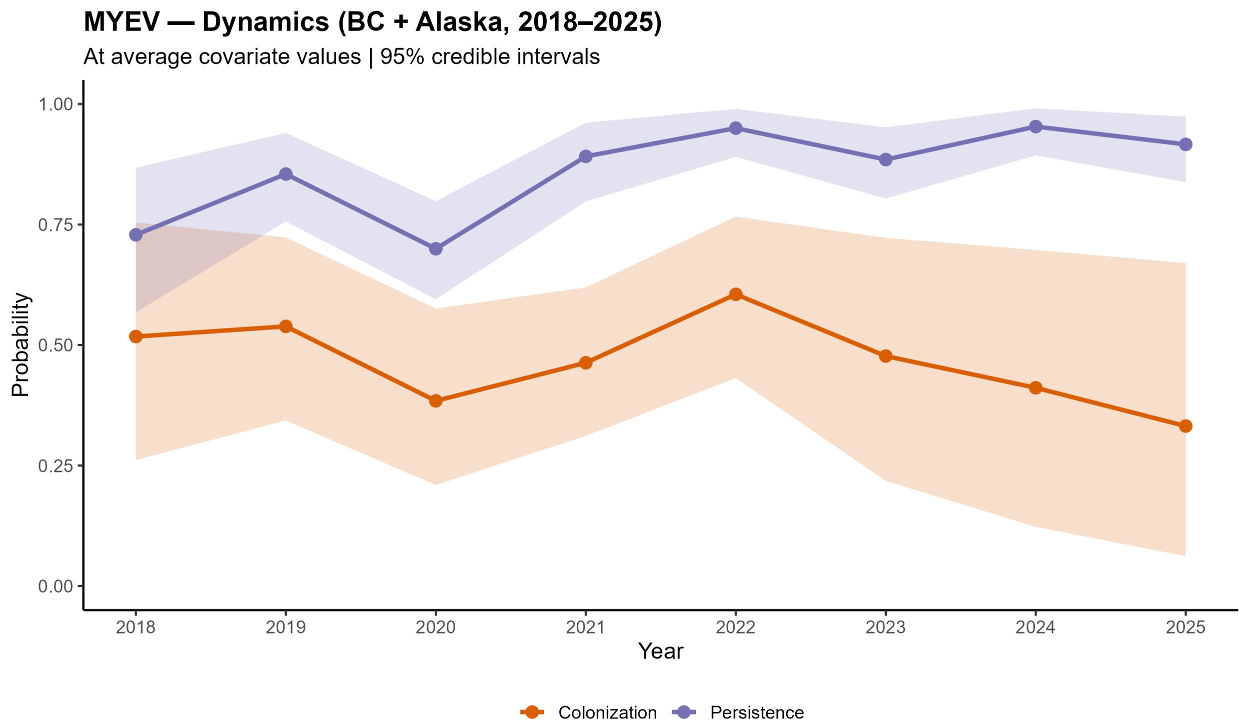

MYCA — Dynamics (BC + Alaska, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

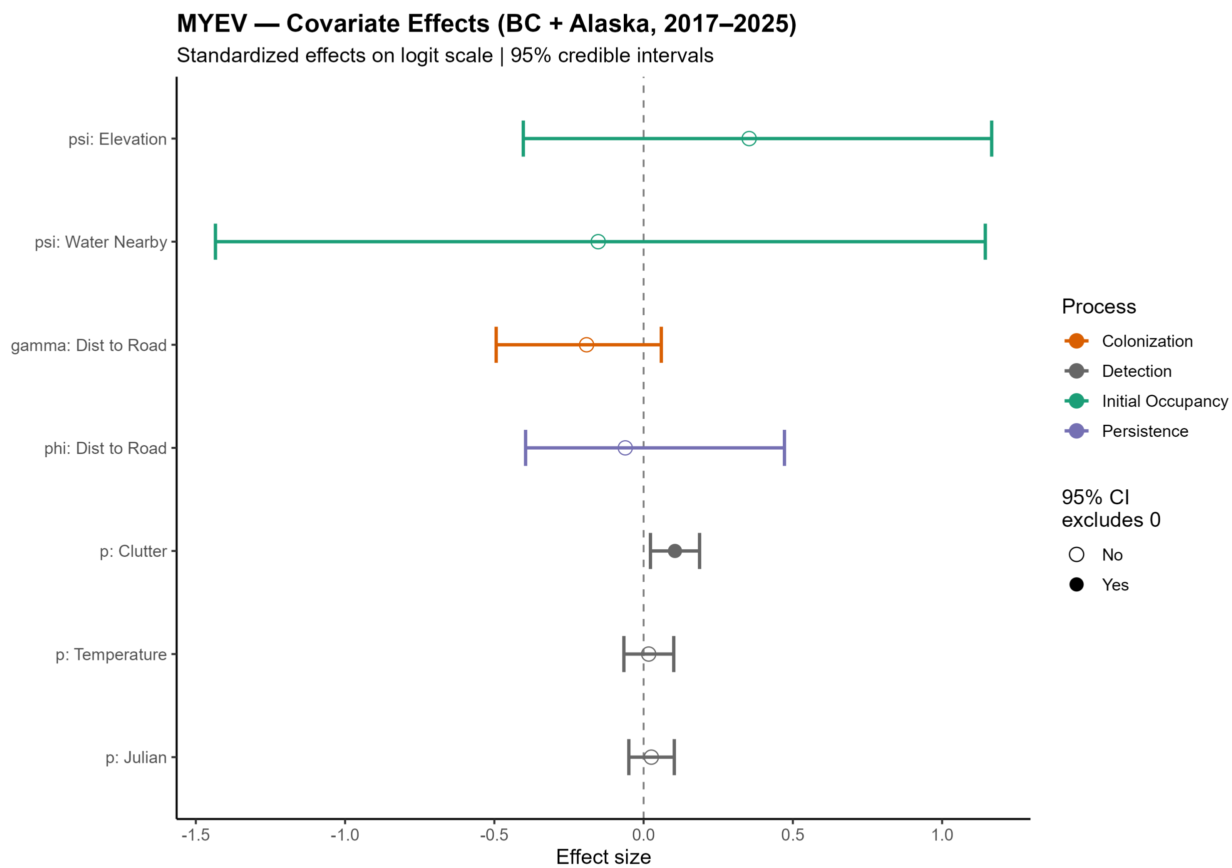

MYCA — Covariate Effects (BC + Alaska, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
phi: Dist to Road
p: Clutter
p: Temperature
p: Julian
-2
-1
0
1
2
3
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

5.3 Long-eared Myotis (*Myotis evotis*)

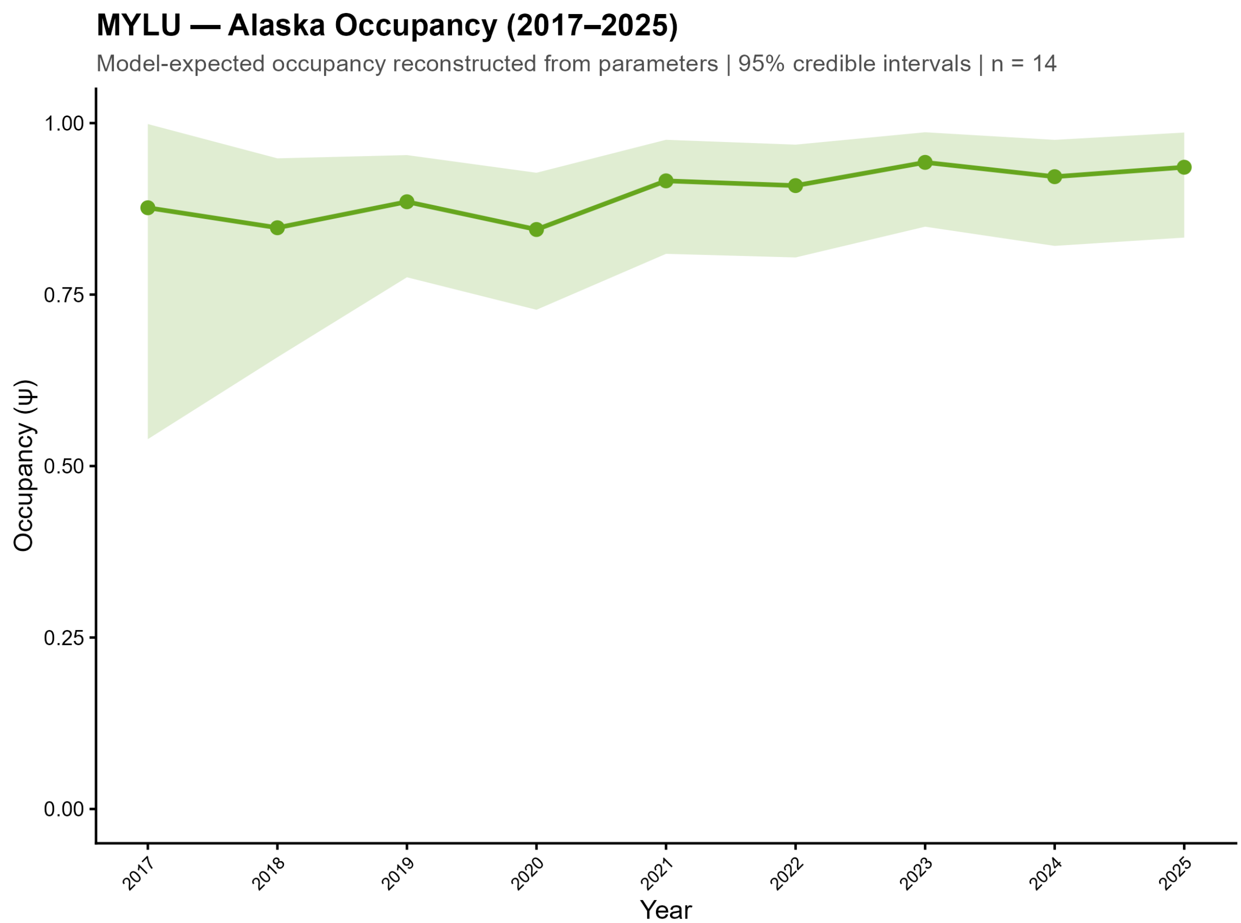


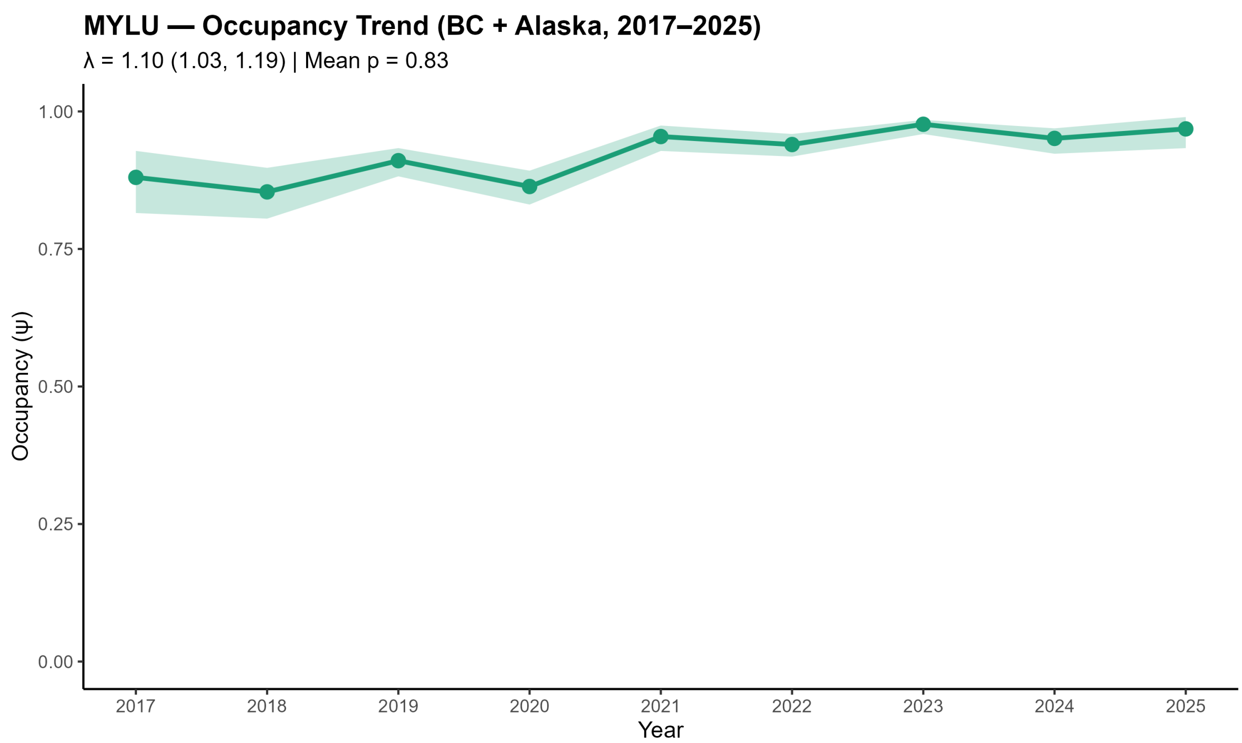


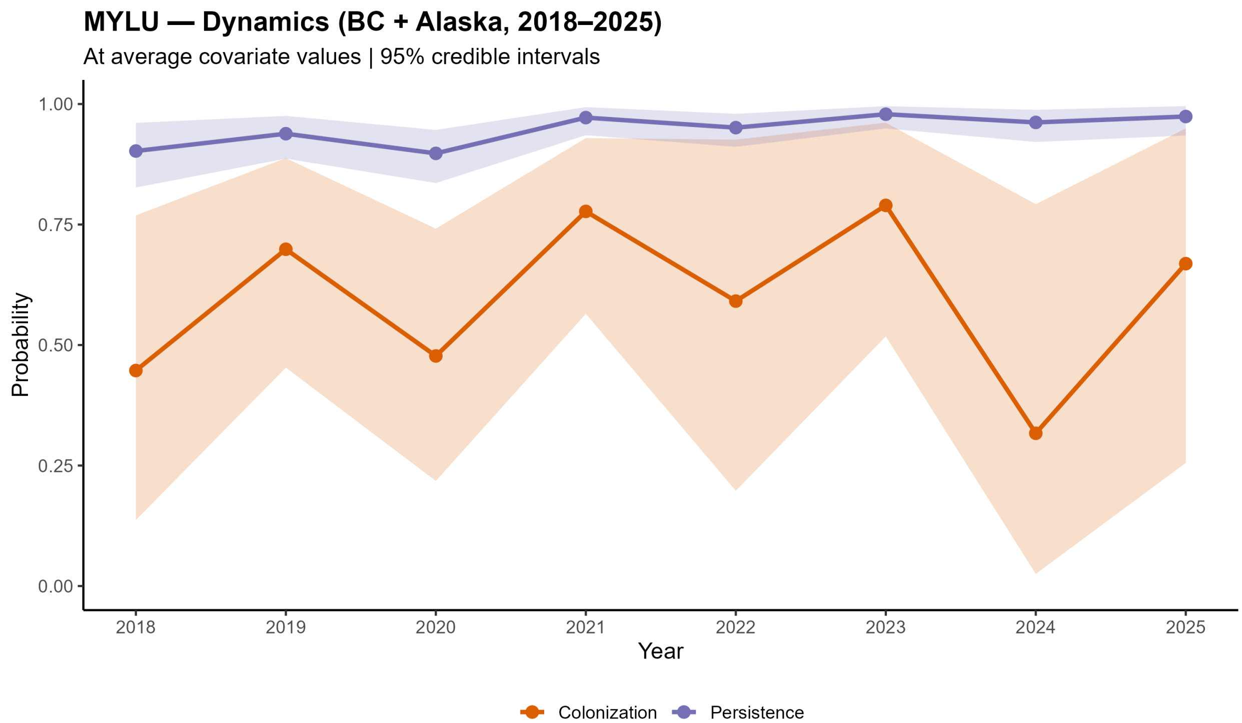

MYEV — Dynamics (BC + Alaska, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


MYEV — Covariate Effects (BC + Alaska, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
phi: Dist to Road
p: Clutter
p: Temperature
p: Julian
-1.5
-1.0
-0.5
0.0
0.5
1.0
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

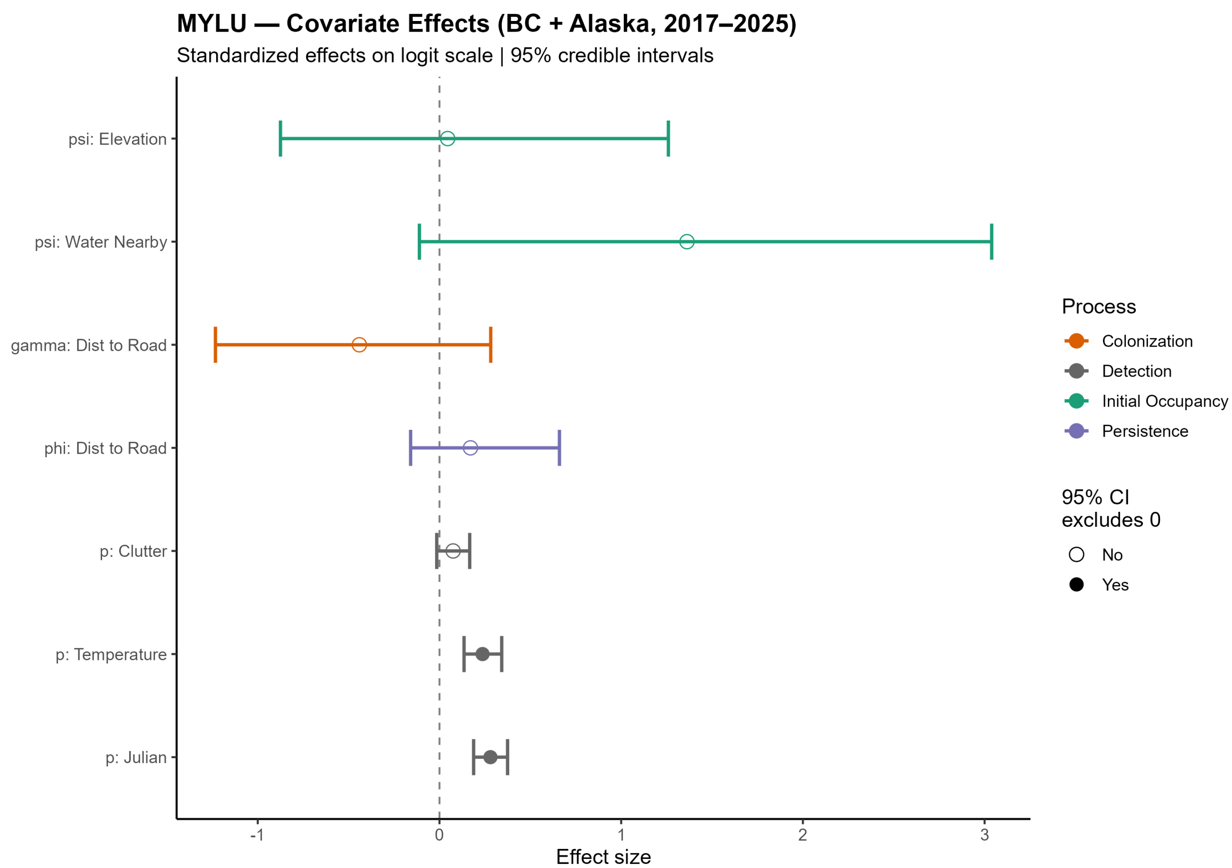
5.4 Little Brown Myotis (*Myotis lucifugus*)





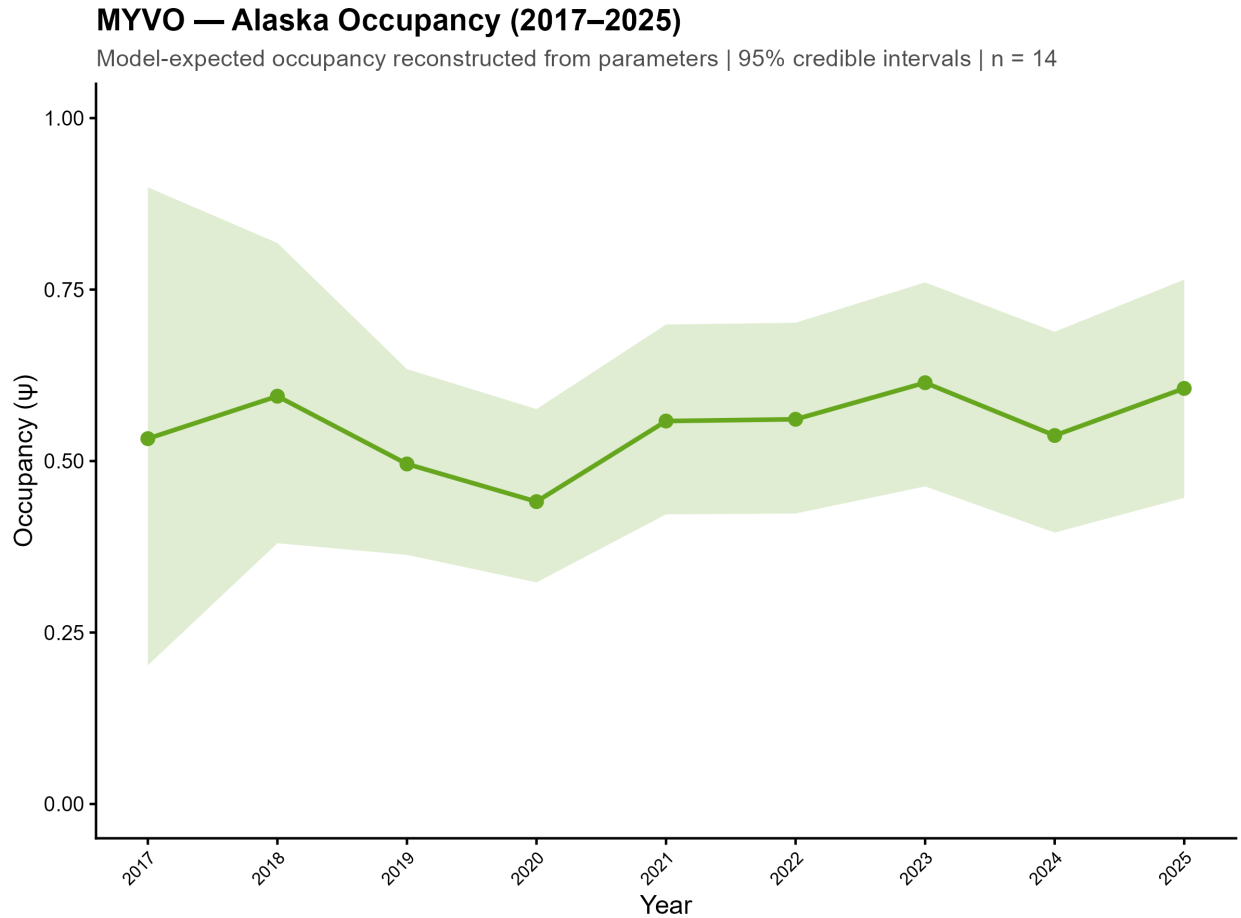


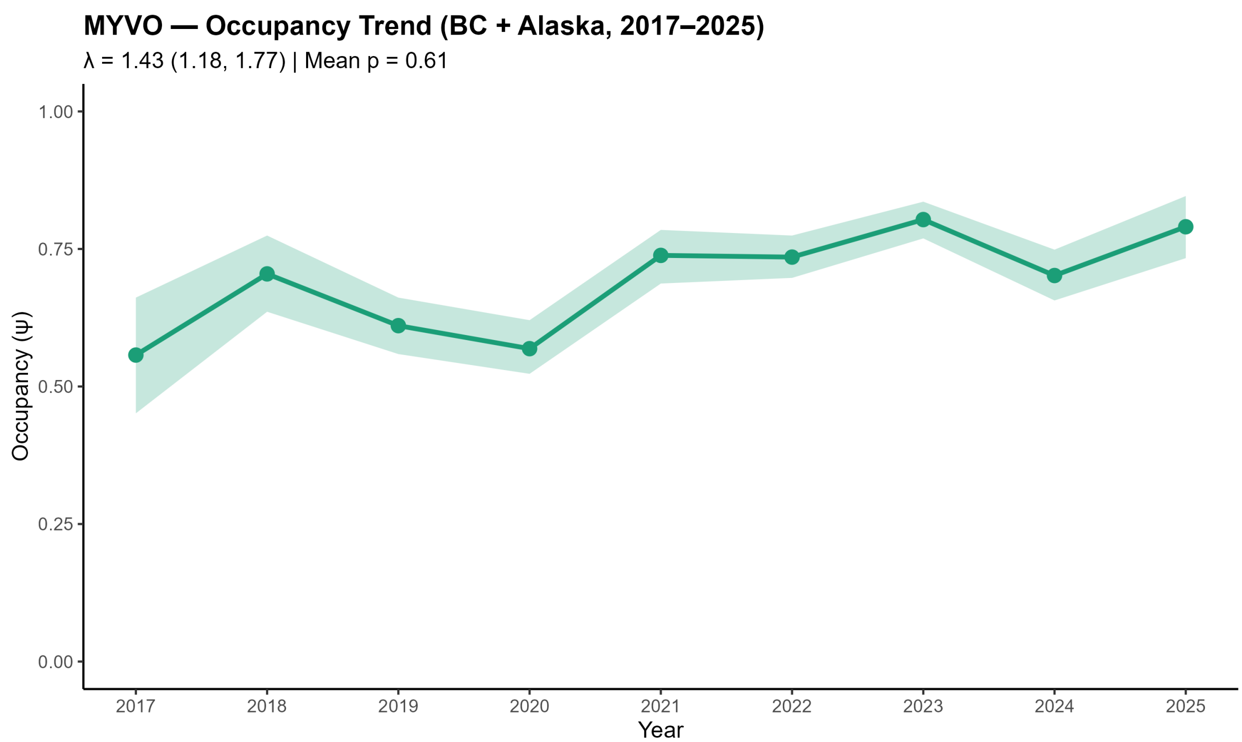
MYLU — Dynamics (BC + Alaska, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

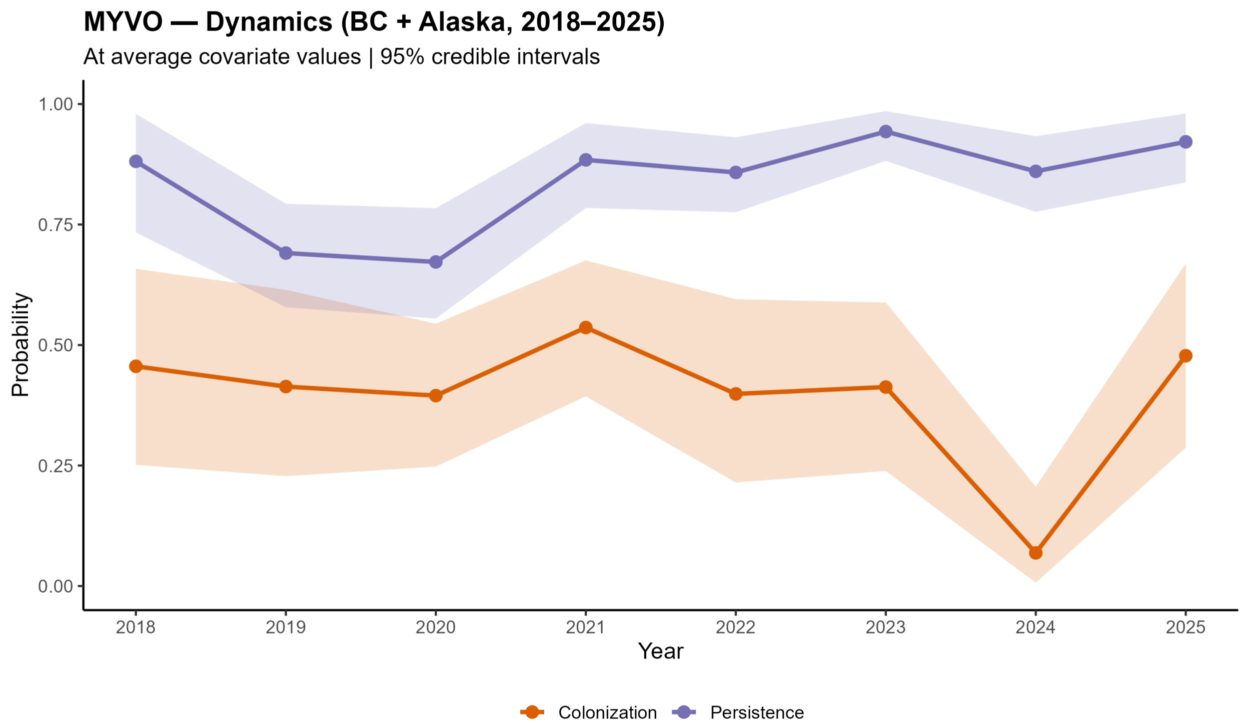


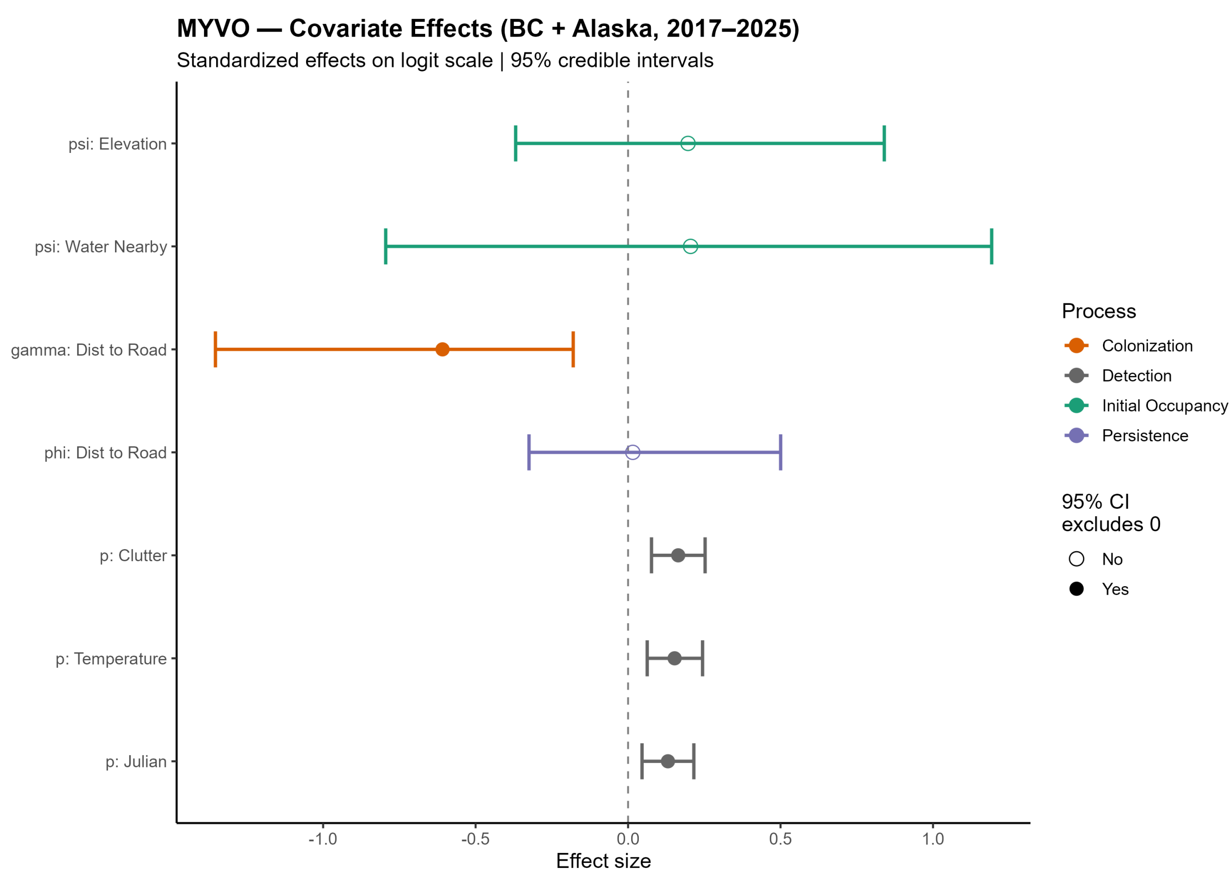
MYLU — Covariate Effects (BC + Alaska, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
phi: Dist to Road
p: Clutter
p: Temperature
p: Julian
-1
0
1
2
3
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

5.5 Long-legged Myotis (*Myotis volans*)





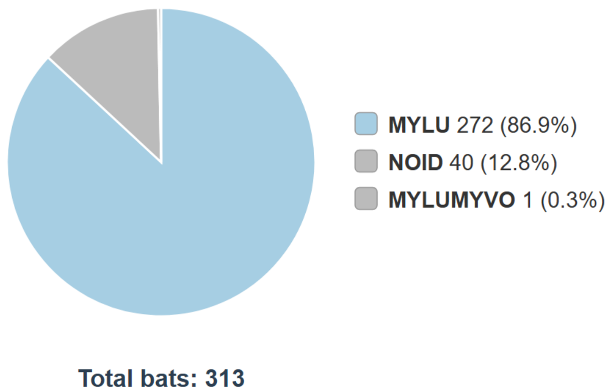

MYVO — Dynamics (BC + Alaska, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


MYVO — Covariate Effects (BC + Alaska, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
phi: Dist to Road
p: Clutter
p: Temperature
p: Julian
-1.0
-0.5
0.0
0.5
1.0
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

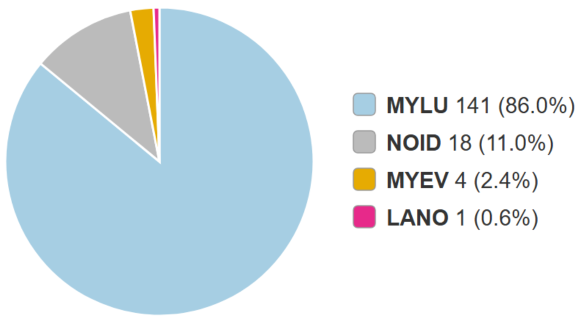
6 Appendix B

Species detection summaries for each monitoring site.

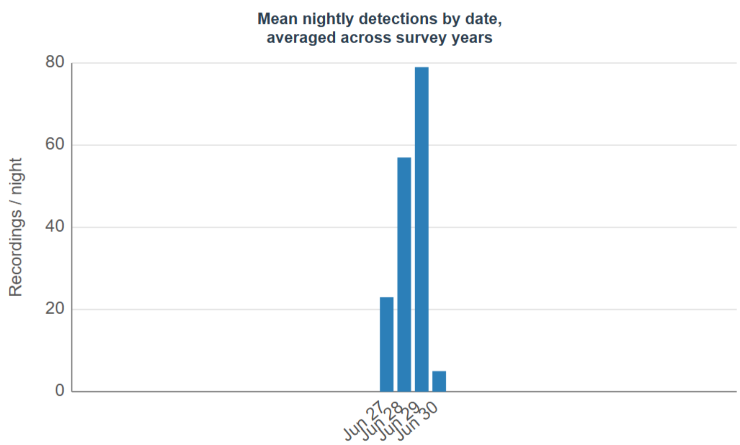
Basher Lake



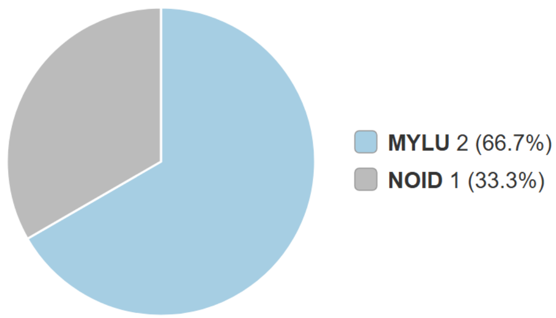
doube2



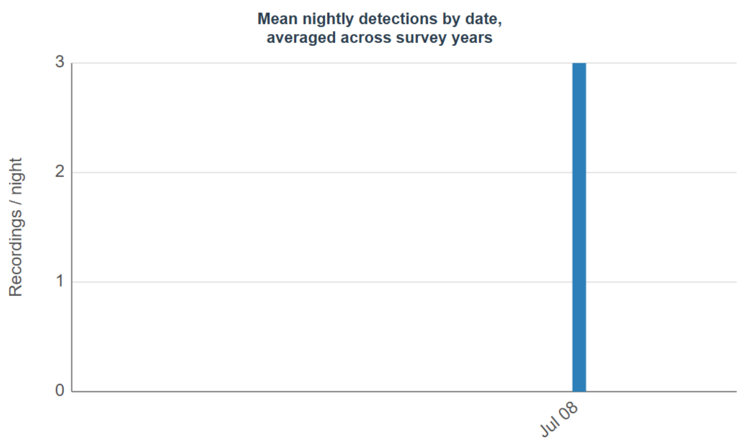
Total bats: 164



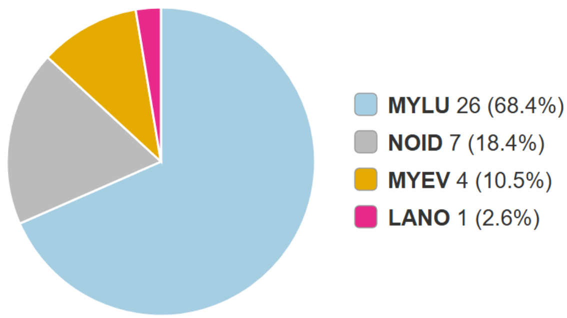
doufr1



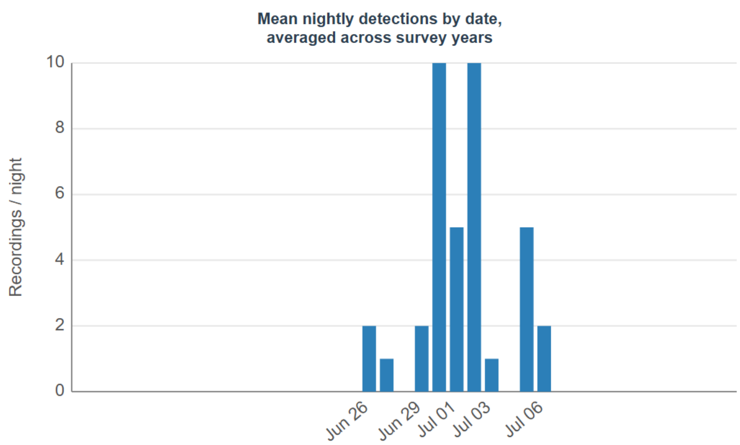
Total bats: 3



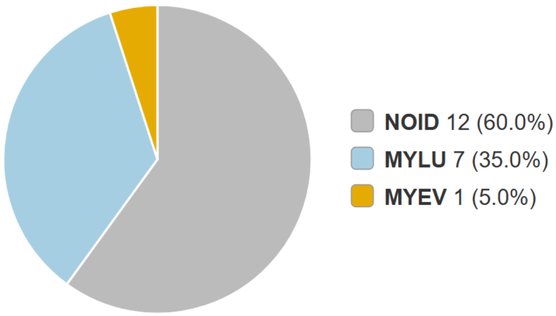
doums1



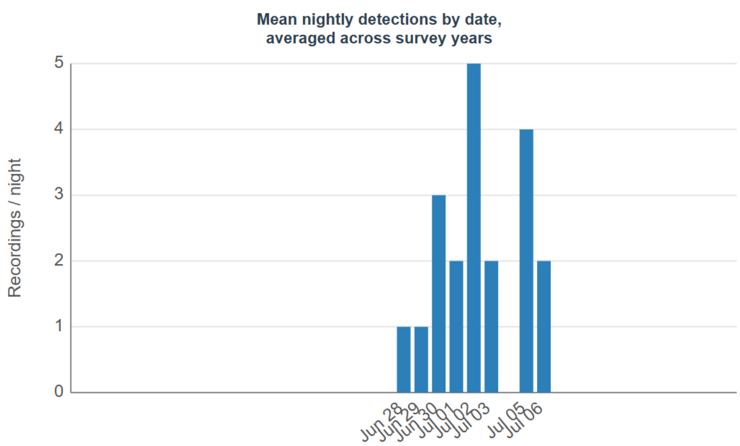
Total bats: 38



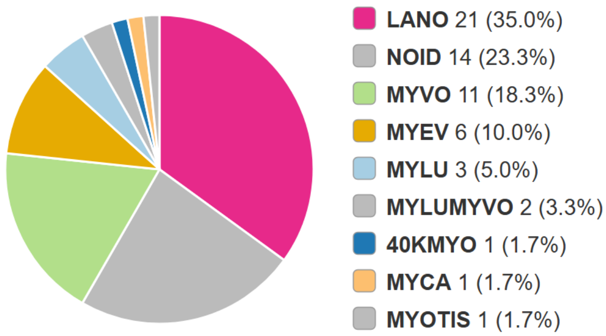
dourp1



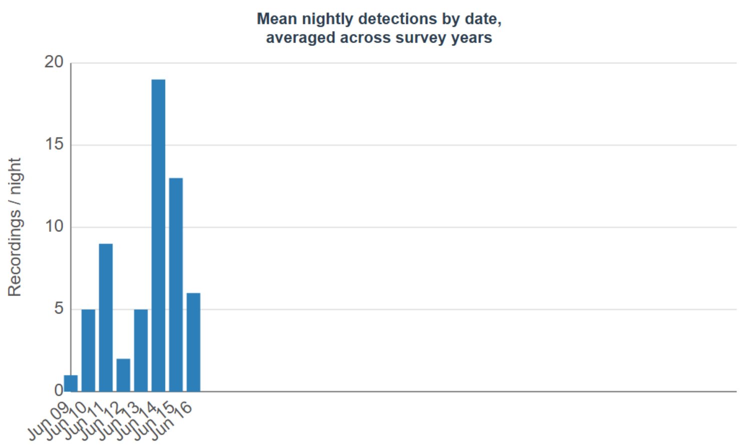
Total bats: 20



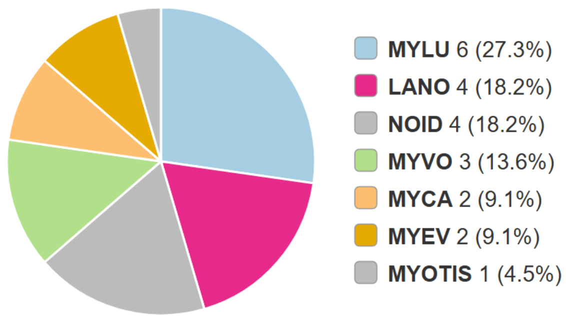
foofr2



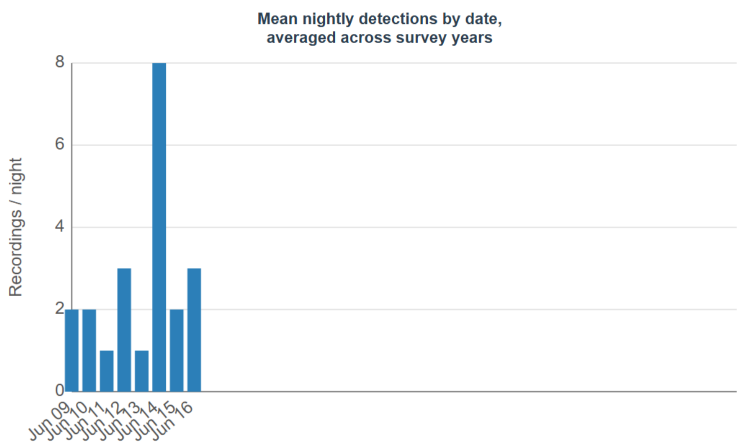
Total bats: 60



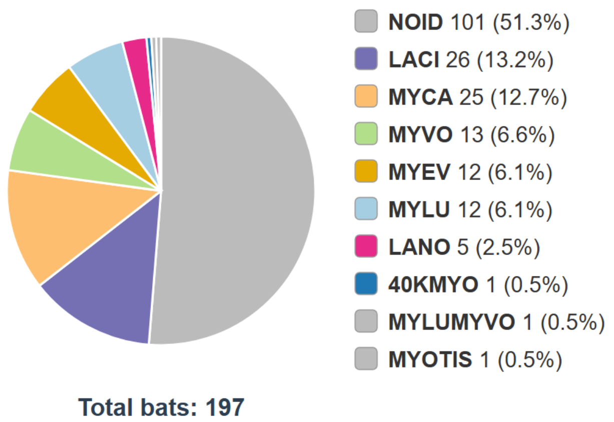
fooms1



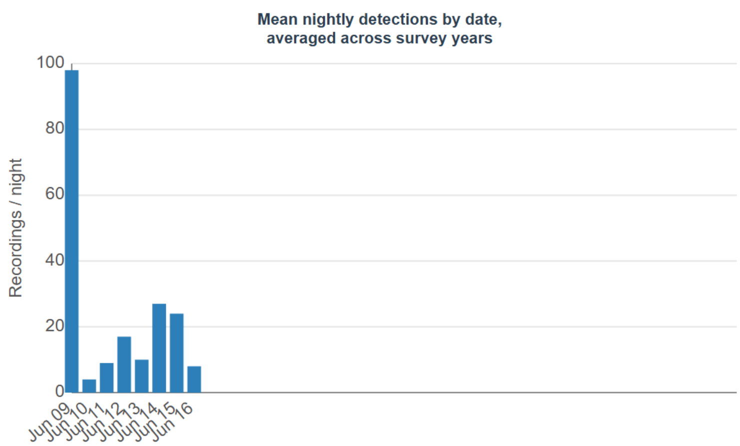
Total bats: 22



foorp1

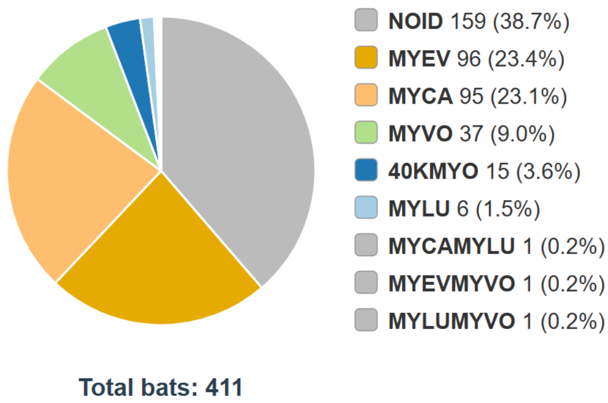


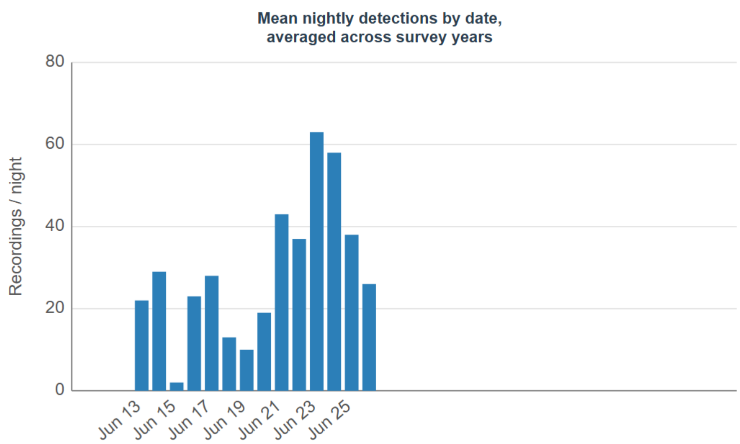
NOID 101 (51.3%)
LACI 26 (13.2%)
MYCA 25 (12.7%)
MYVO 13 (6.6%)
MYEV 12 (6.1%)
MYLU 12 (6.1%)
LANO 5 (2.5%)
40KMYO 1 (0.5%)
MYLUMYVO 1 (0.5%)
MYOTIS 1 (0.5%)
Total bats: 197



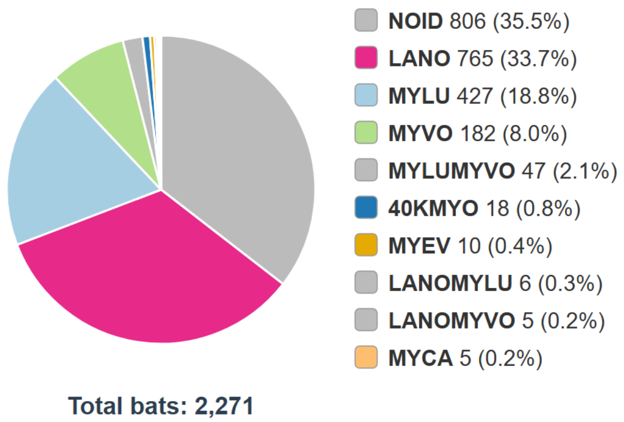
Mean nightly detections by date,
averaged across survey years
Recordings / night
100
80
60
40
20
0
Jun 09
Jun 10
Jun 11
Jun 12
Jun 13
Jun 14
Jun 15
Jun 16

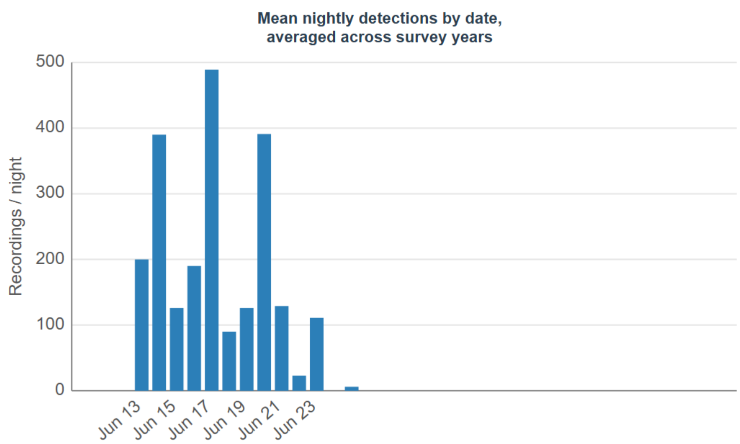
hhlfr1


NOID 159 (38.7%)
MYEV 96 (23.4%)
MYCA 95 (23.1%)
MYVO 37 (9.0%)
40KMYO 15 (3.6%)
MYLU 6 (1.5%)
MYCAMYLU 1 (0.2%)
MYEVMYVO 1 (0.2%)
MYLUMYVO 1 (0.2%)
Total bats: 411

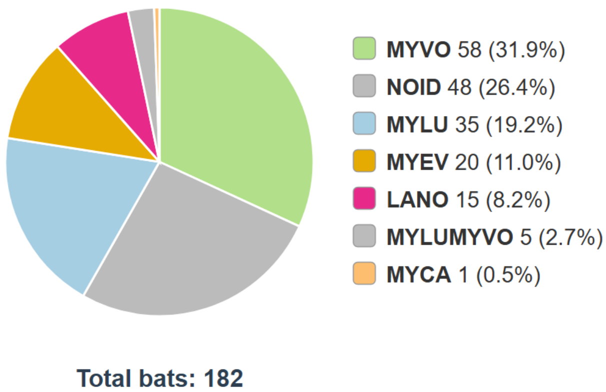

Mean nightly detections by date,
averaged across survey years
Recordings / night
0
20
40
60
80
Jun 13
Jun 15
Jun 17
Jun 19
Jun 21
Jun 23
Jun 25

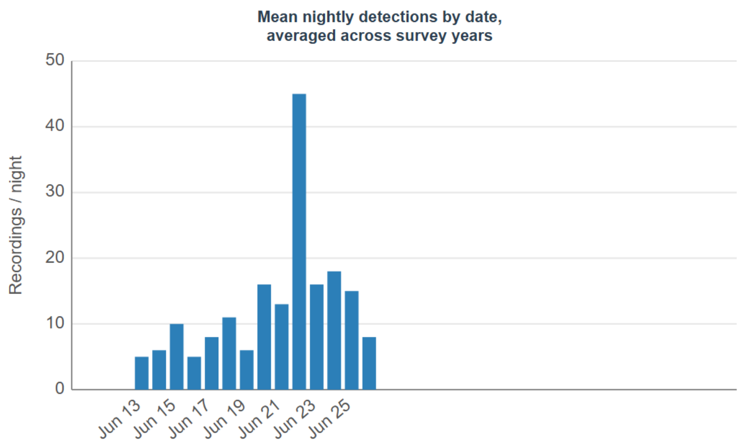
hhllk1


NOID 806 (35.5%)
LANO 765 (33.7%)
MYLU 427 (18.8%)
MYVO 182 (8.0%)
MYLUMYVO 47 (2.1%)
40KMYO 18 (0.8%)
MYEV 10 (0.4%)
LANOMYLU 6 (0.3%)
LANOMYVO 5 (0.2%)
MYCA 5 (0.2%)
Total bats: 2,271

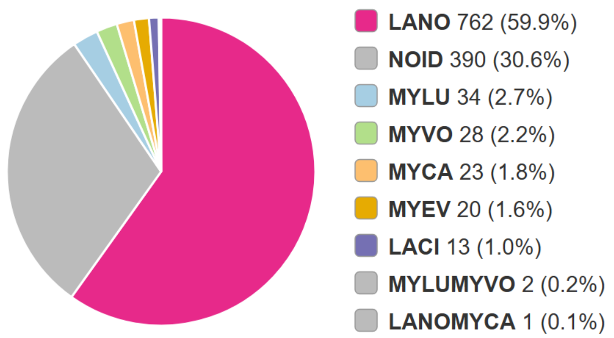

Mean nightly detections by date,
averaged across survey years
Recordings / night
500
400
300
200
100
0
Jun 13
Jun 15
Jun 17
Jun 19
Jun 21
Jun 23

hhlms1

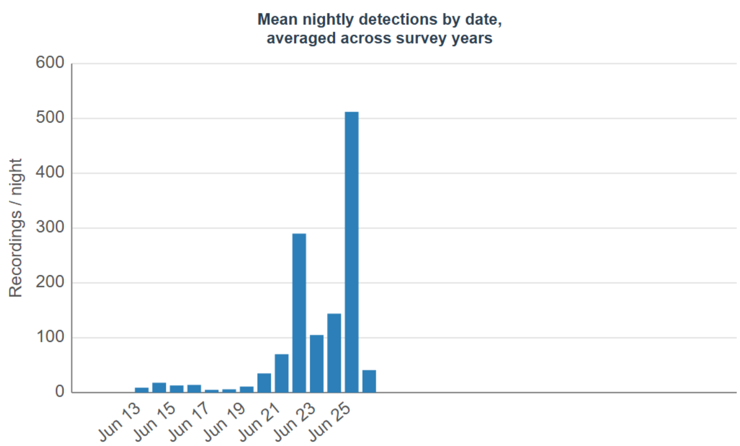

MYVO 58 (31.9%)
NOID 48 (26.4%)
MYLU 35 (19.2%)
MYEV 20 (11.0%)
LANO 15 (8.2%)
MYLUMYVO 5 (2.7%)
MYCA 1 (0.5%)
Total bats: 182


Mean nightly detections by date,
averaged across survey years
Recordings / night
0
10
20
30
40
50
Jun 13
Jun 15
Jun 17
Jun 19
Jun 21
Jun 23
Jun 25

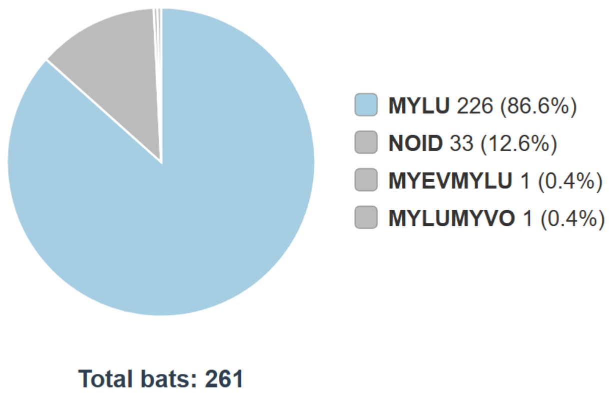
hhlrp1

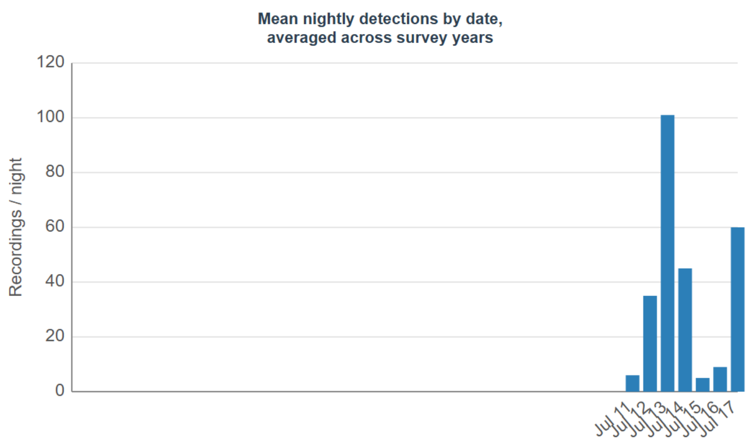


Total bats: 1,273

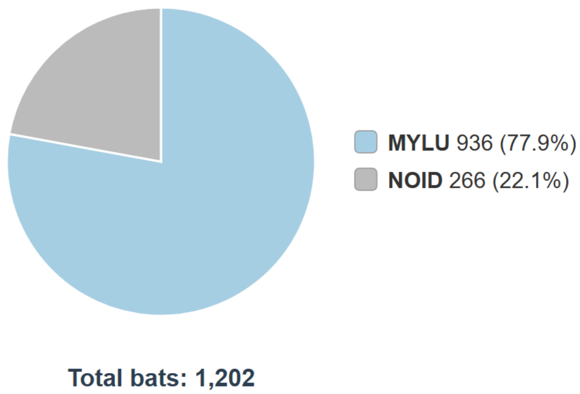


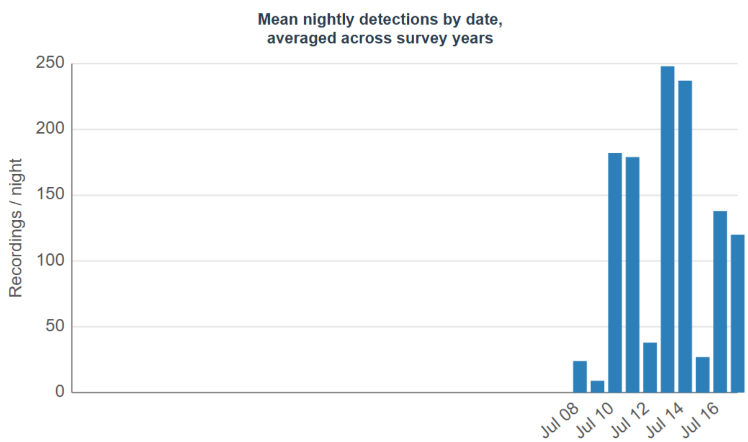
Jewel Lake


MYLU 226 (86.6%)
NOID 33 (12.6%)
MYEVMYLU 1 (0.4%)
MYLUMYVO 1 (0.4%)
Total bats: 261

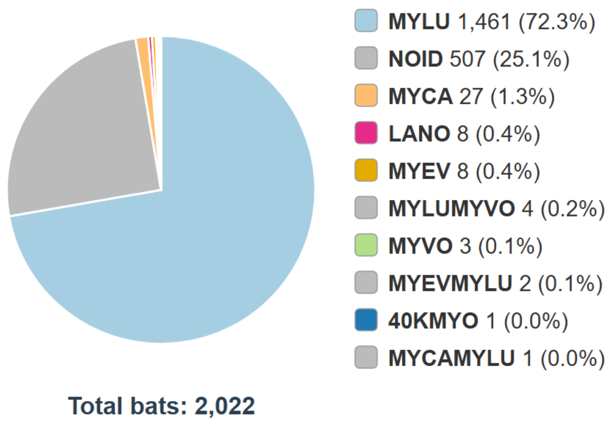

Mean nightly detections by date,
averaged across survey years
Recordings / night
0
20
40
60
80
100
120

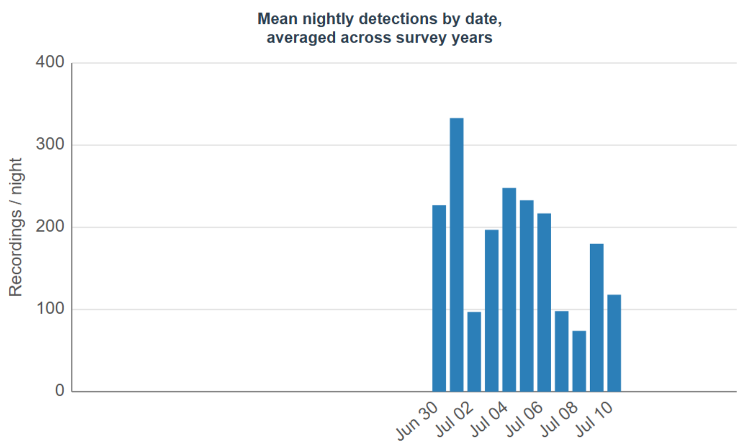
Little Campbell Lake



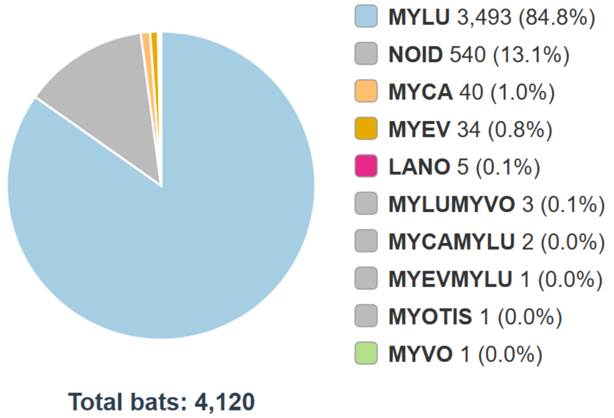


mvafr1

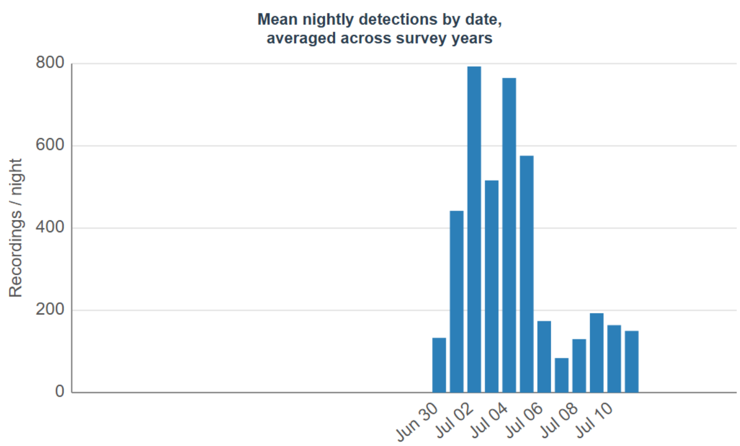

MYLU 1,461 (72.3%)
NOID 507 (25.1%)
MYCA 27 (1.3%)
LANO 8 (0.4%)
MYEV 8 (0.4%)
MYLUMYVO 4 (0.2%)
MYVO 3 (0.1%)
MYEVMYLU 2 (0.1%)
40KMYO 1 (0.0%)
MYCAMYLU 1 (0.0%)
Total bats: 2,022


Mean nightly detections by date,
averaged across survey years
Recordings / night
400
300
200
100
0
Jun 30
Jul 02
Jul 04
Jul 06
Jul 08
Jul 10

mvapd1

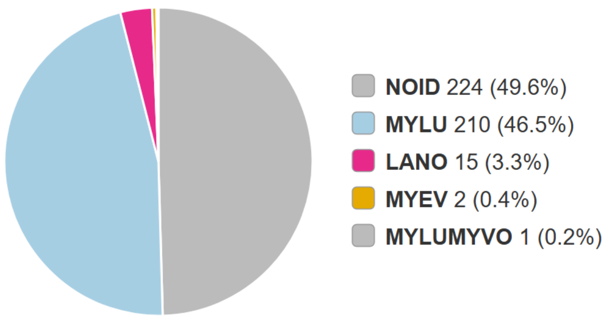


MYLU 3,493 (84.8%)
NOID 540 (13.1%)
MYCA 40 (1.0%)
MYEV 34 (0.8%)
LANO 5 (0.1%)
MYLUMYVO 3 (0.1%)
MYCAMYLU 2 (0.0%)
MYEVMYLU 1 (0.0%)
MYOTIS 1 (0.0%)
MYVO 1 (0.0%)
Total bats: 4,120

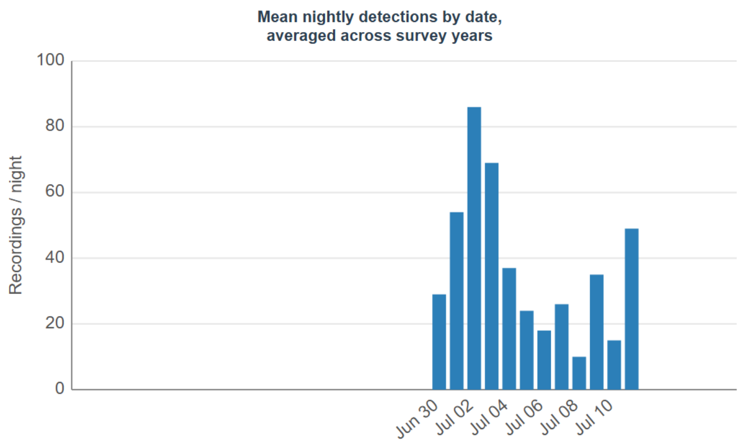


Mean nightly detections by date,
averaged across survey years
Recordings / night
800
600
400
200
0
Jun 30
Jul 02
Jul 04
Jul 06
Jul 08
Jul 10

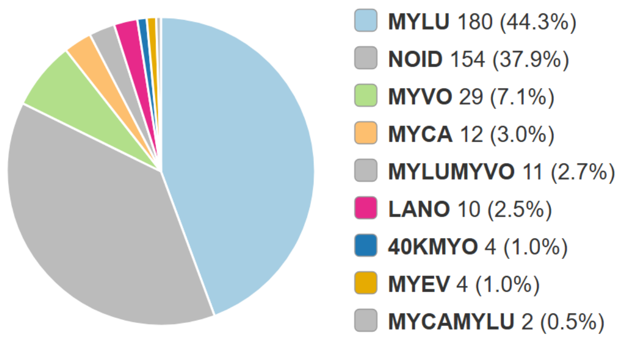
mvawt2



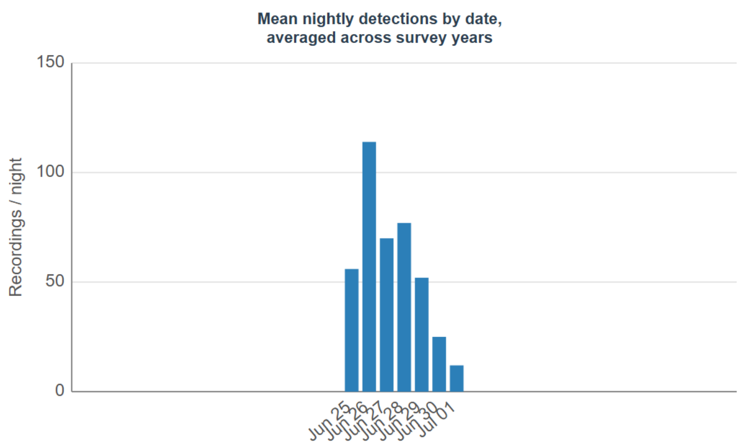
Total bats: 452



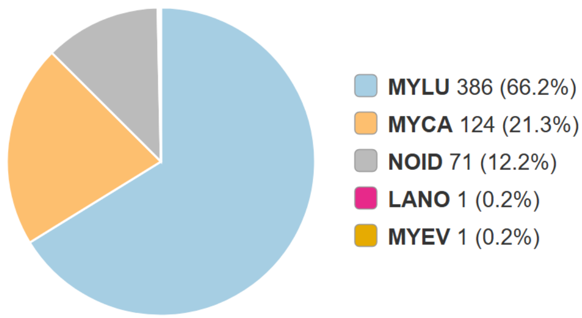
stkbe3



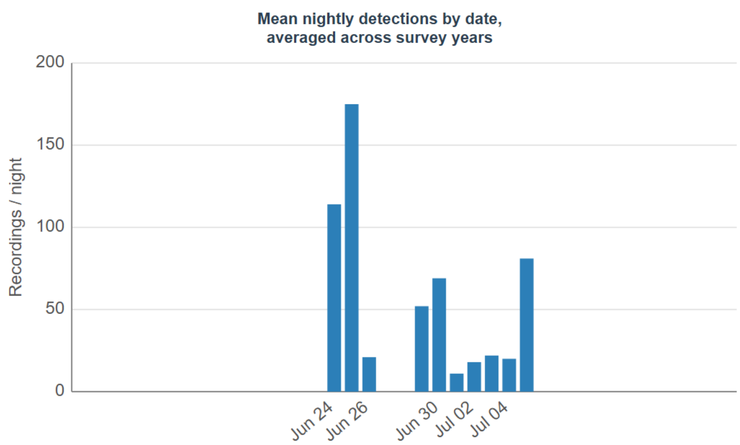
Total bats: 406



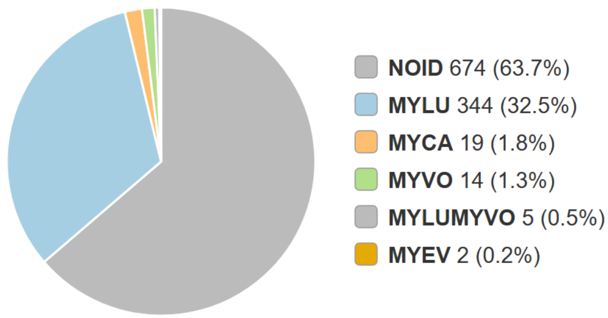
stkms2



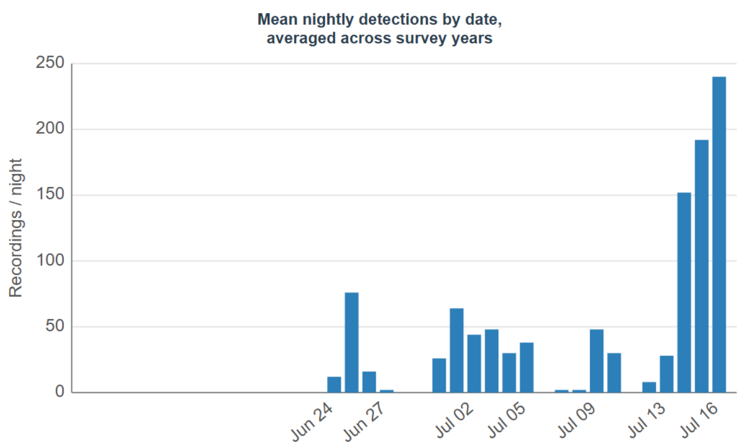
Total bats: 583



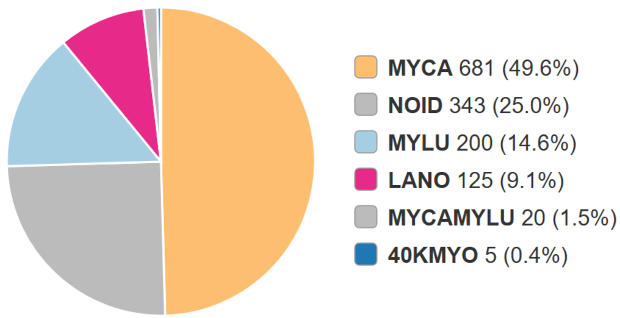
stkpd1



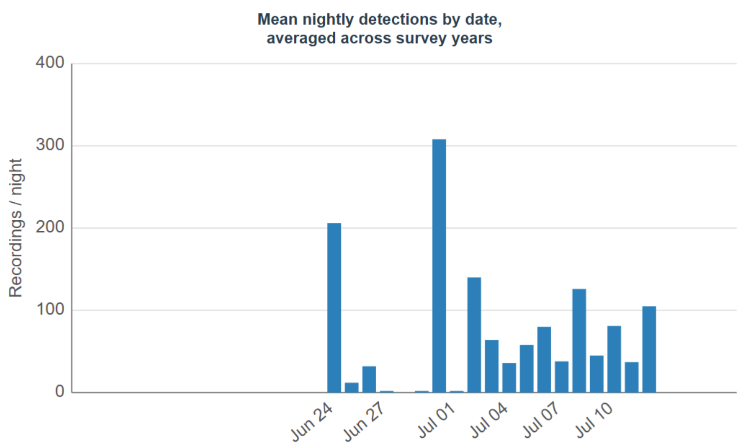
Total bats: 1,058



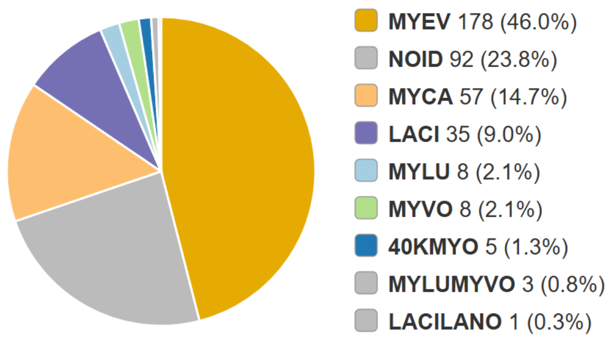
stkrp2



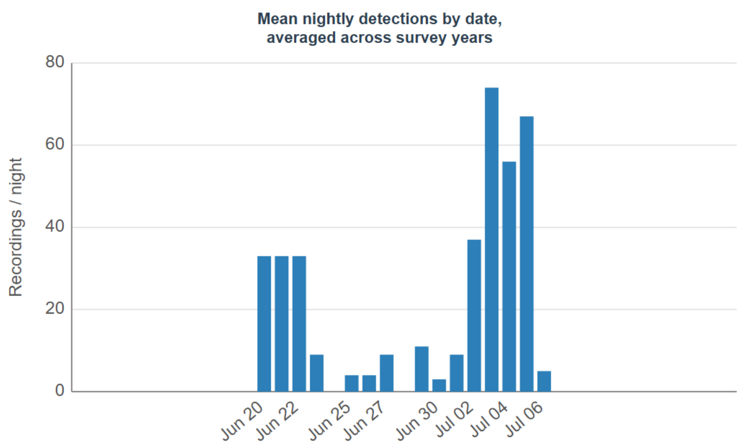
Total bats: 1,374



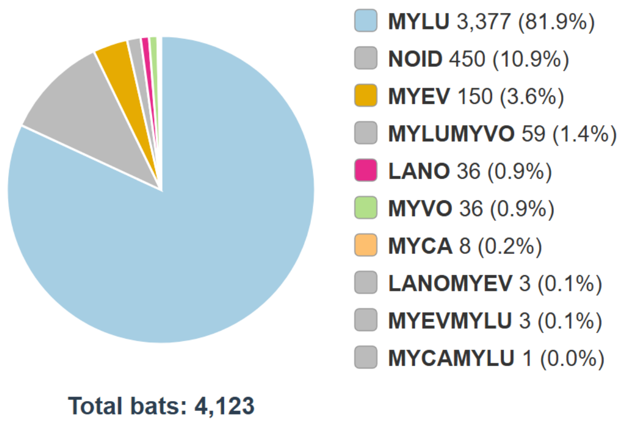
tllfr2



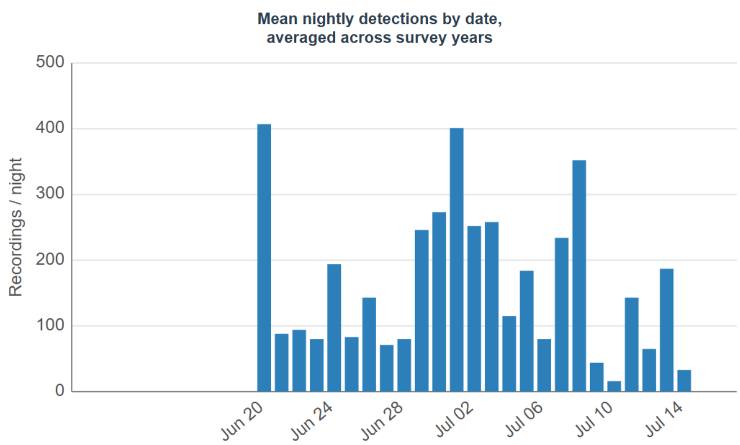
Total bats: 387



tIllk2

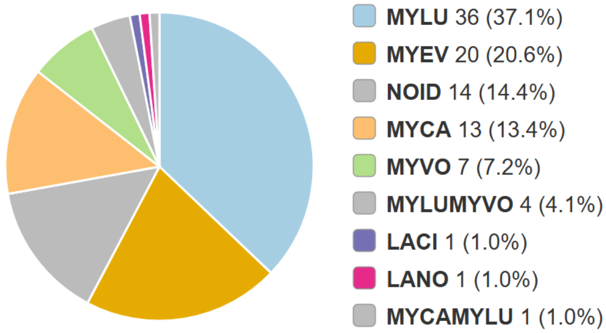


MYLU 3,377 (81.9%)
NOID 450 (10.9%)
MYEV 150 (3.6%)
MYLUMYVO 59 (1.4%)
LANO 36 (0.9%)
MYVO 36 (0.9%)
MYCA 8 (0.2%)
LANOMYEV 3 (0.1%)
MYEVMYLU 3 (0.1%)
MYCAMYLU 1 (0.0%)
Total bats: 4,123

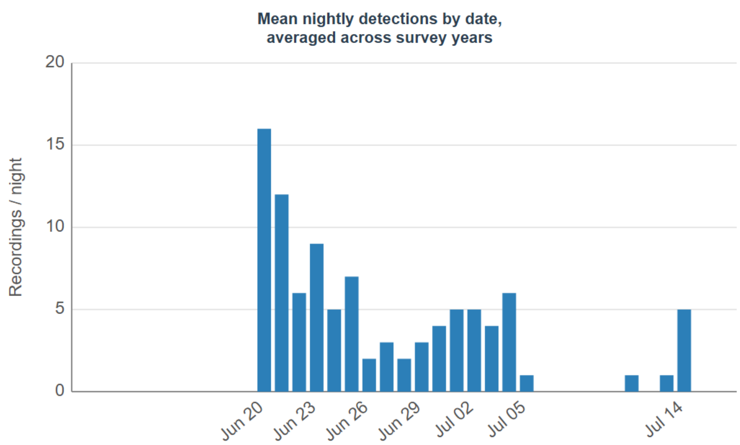


Mean nightly detections by date,
averaged across survey years
Recordings / night
500
400
300
200
100
0
Jun 20
Jun 24
Jun 28
Jul 02
Jul 06
Jul 10
Jul 14

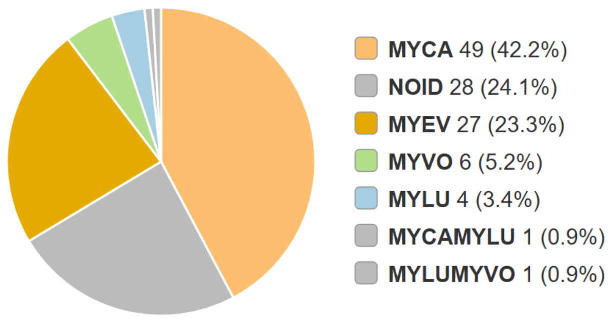
tllms1



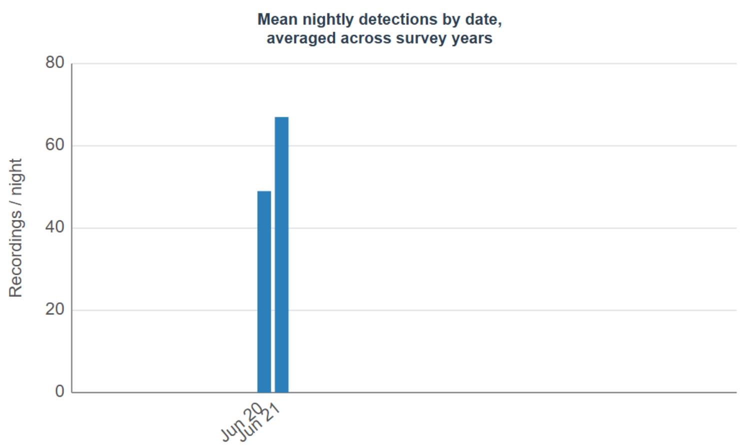
Total bats: 97



tllrp2



Total bats: 116



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